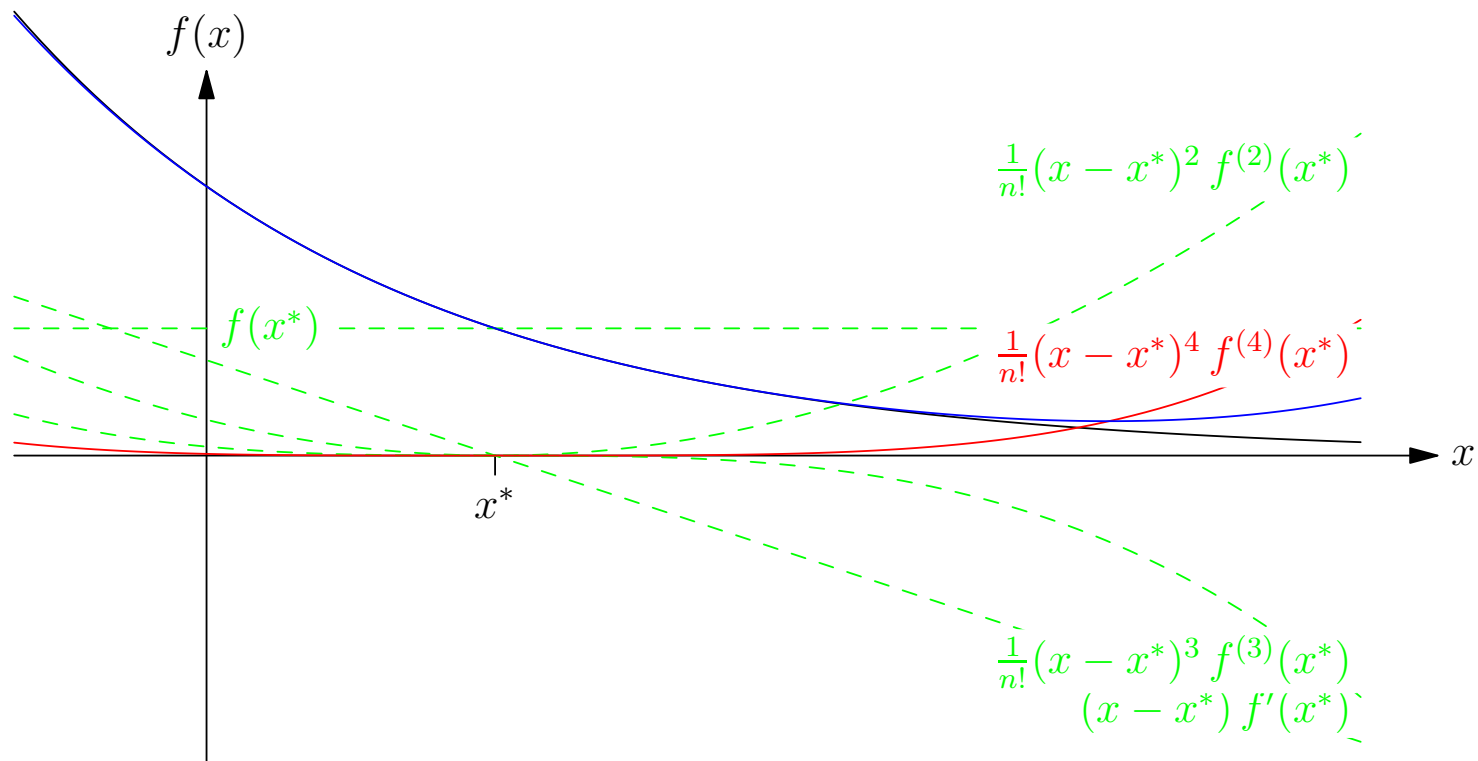


Mathematics of Machine Learning

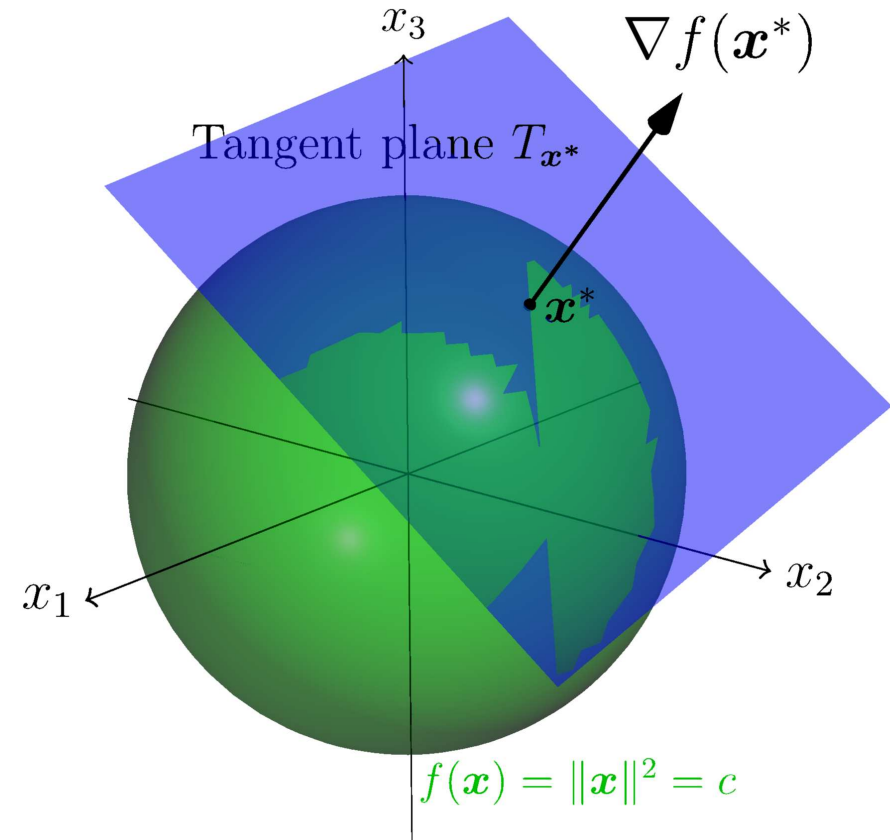
Taylor's Expansion



Converge, residue, high-dimensions, Hessian

Outline

1. **1-D Taylor Expansion**
2. High Dimensions
3. Hessians



Approximating Functions

- In much of machine learning we are trying to approximate functions
- There are many different ways to do this
- Depending on your application some are much better than others
- All function approximation makes some assumptions about the regularity of the function they are approximating
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- Suppose we wanted to approximate a function $f(x)$ around a point x^* using a polynomial

$$f(x) \approx a_0 + a_1(x - x^*) + a_2(x - x^*)^2 + \dots + a_n(x - x^*)^n$$

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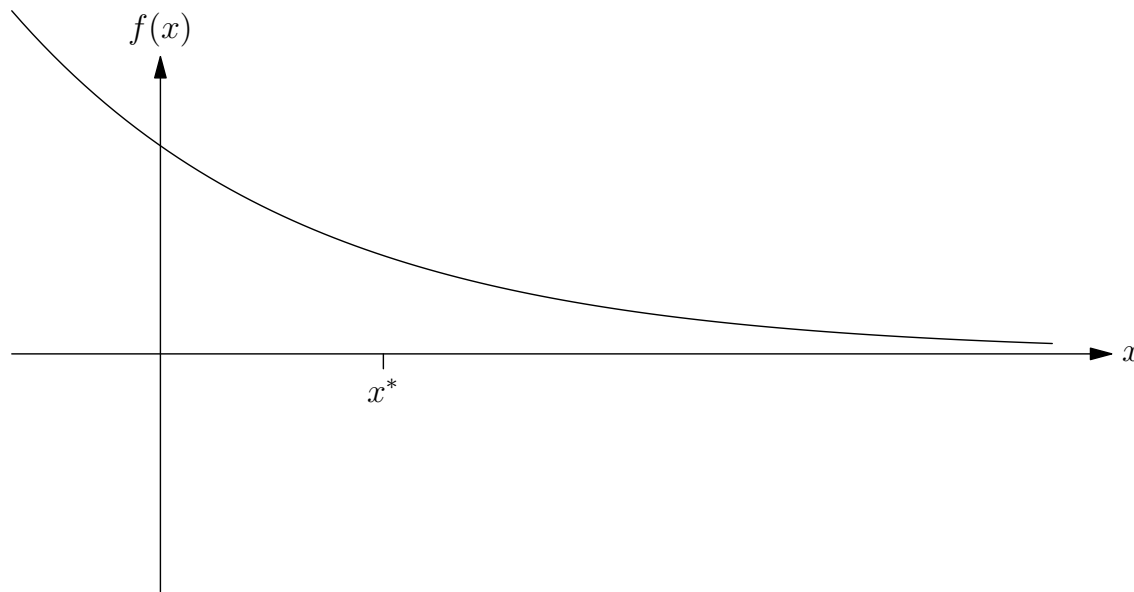
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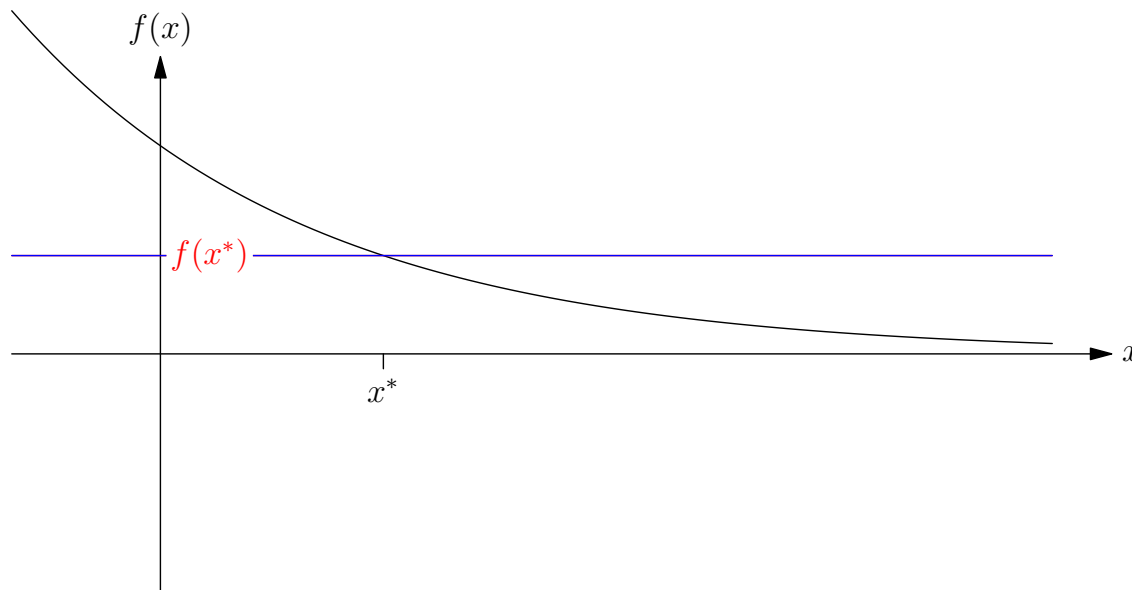


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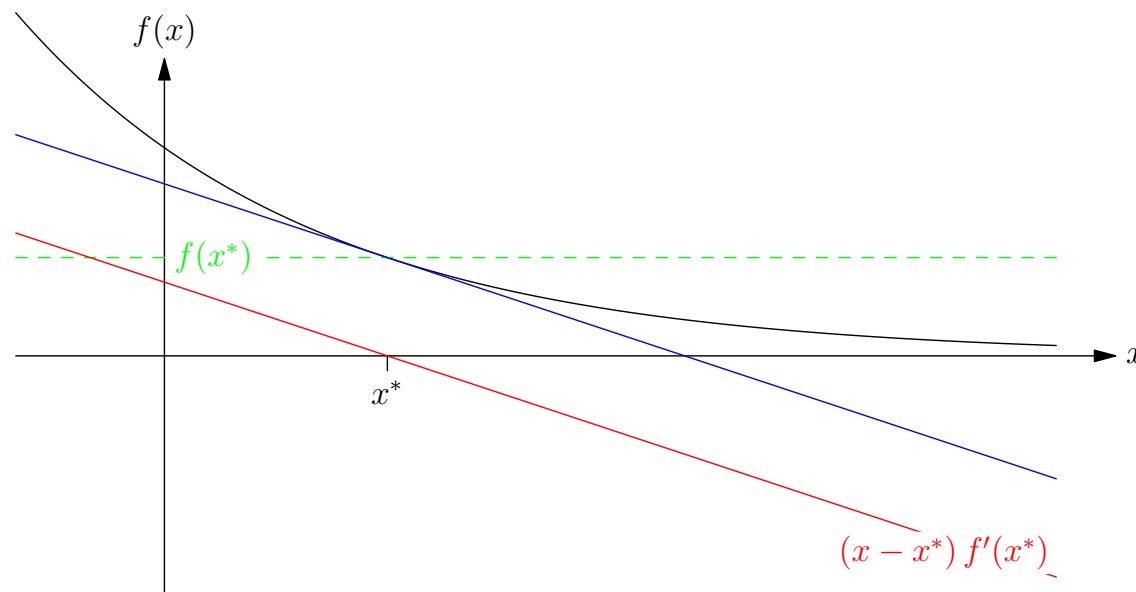


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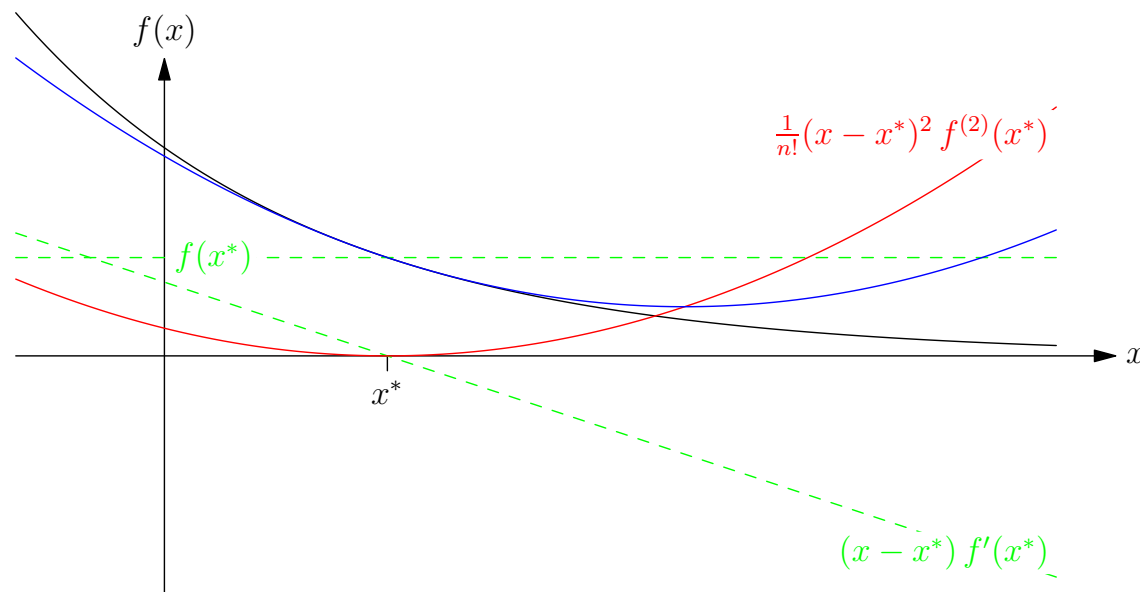


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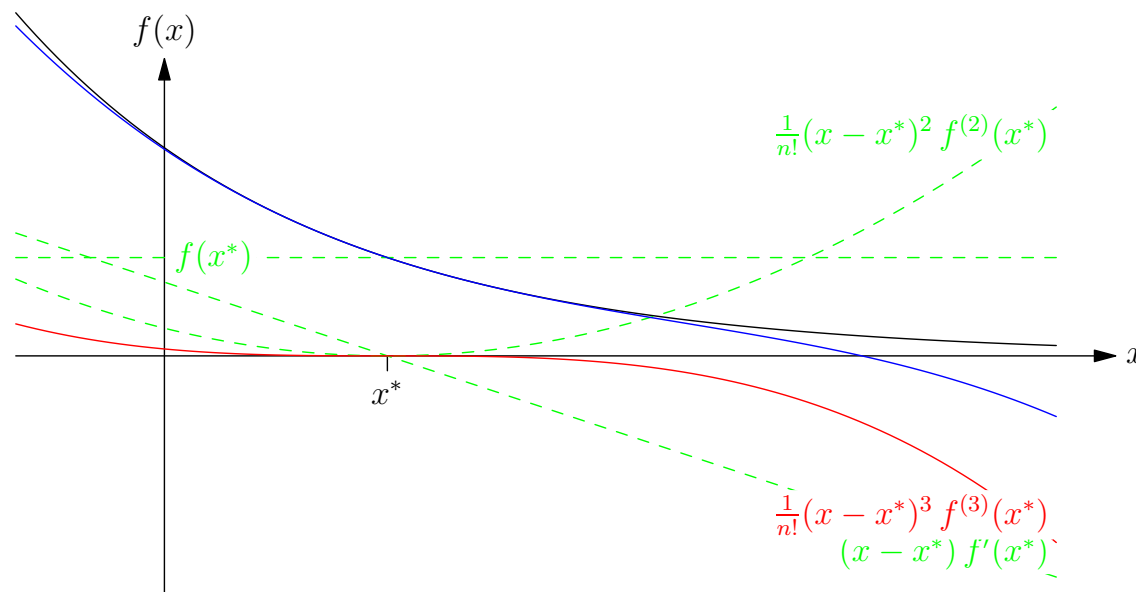


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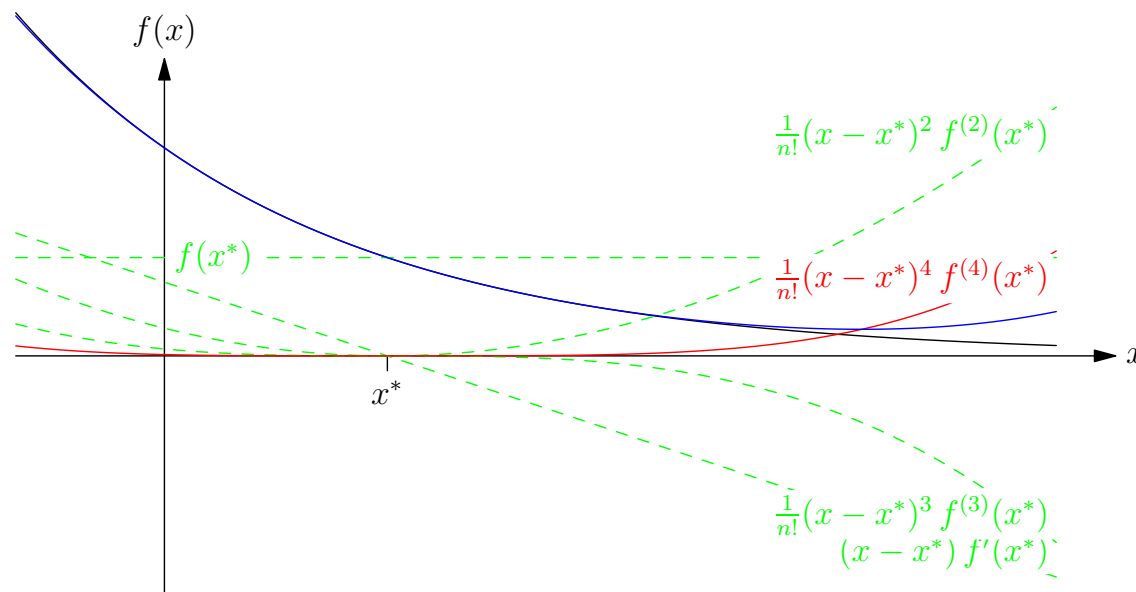


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Convergence

- As we add more and more terms our local approximation becomes more accurate
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- However, polynomials cannot capture functions that diverge for finite numbers, e.g.

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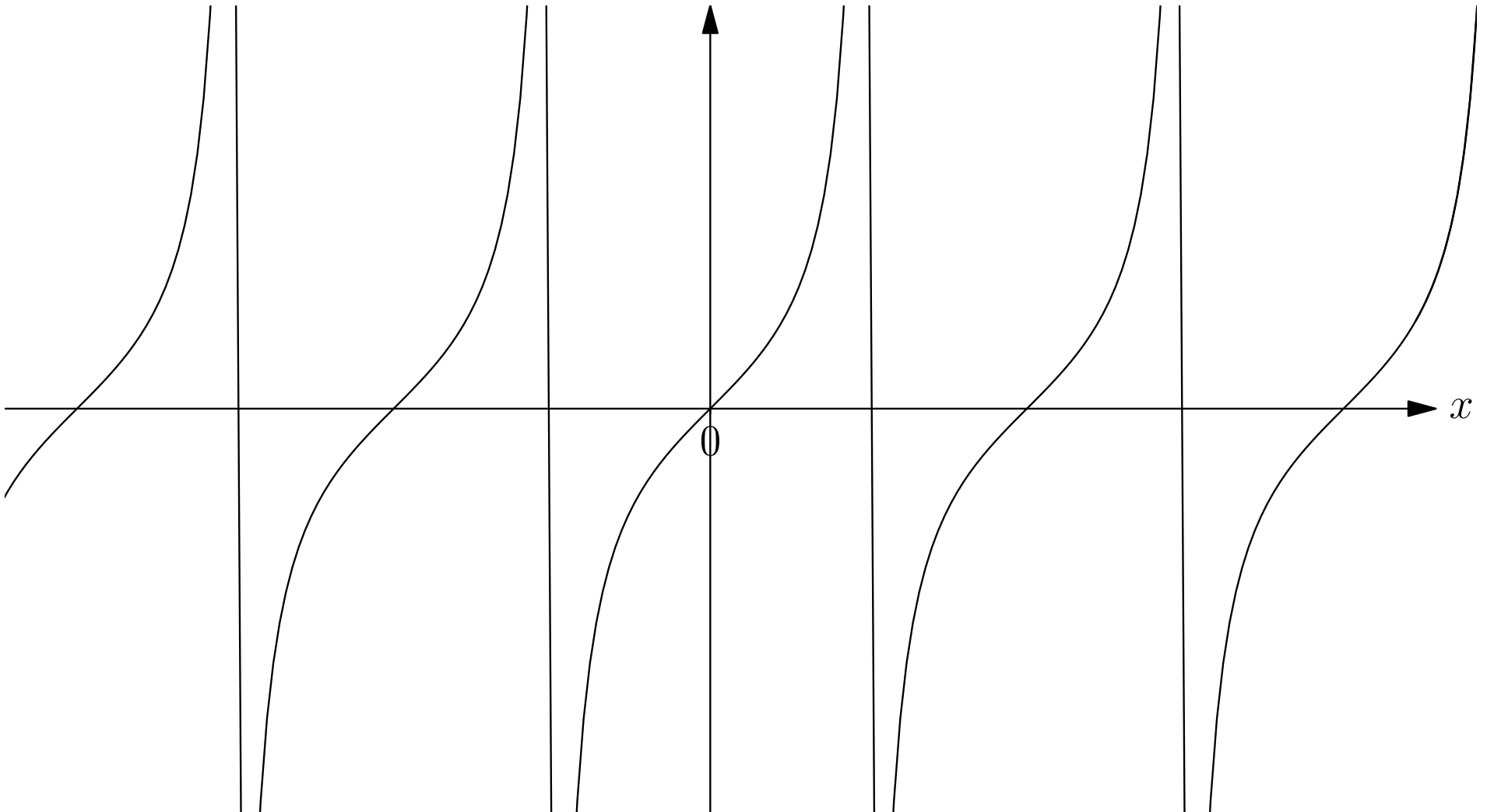
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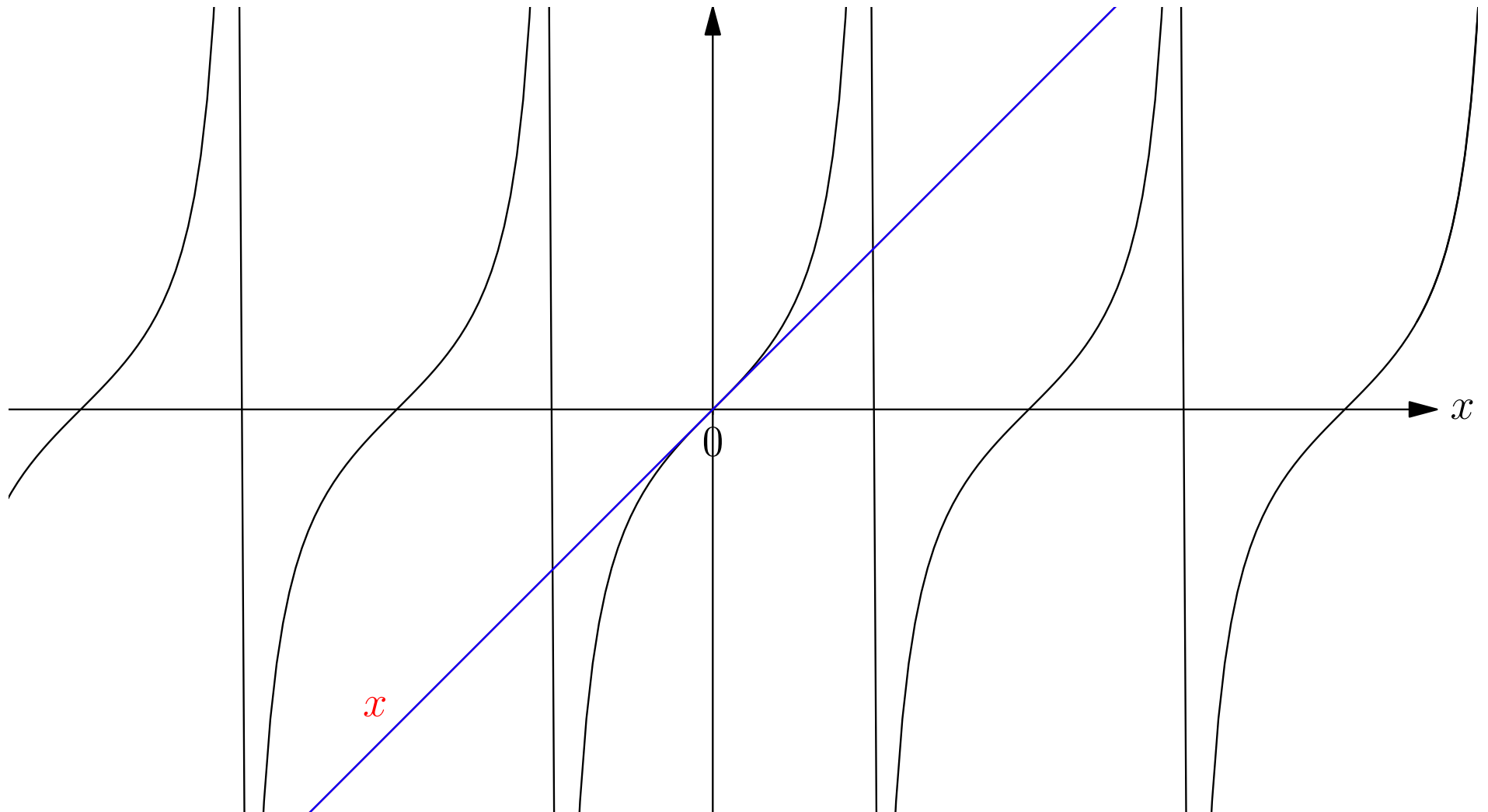
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- This leads to a loss of global convergence

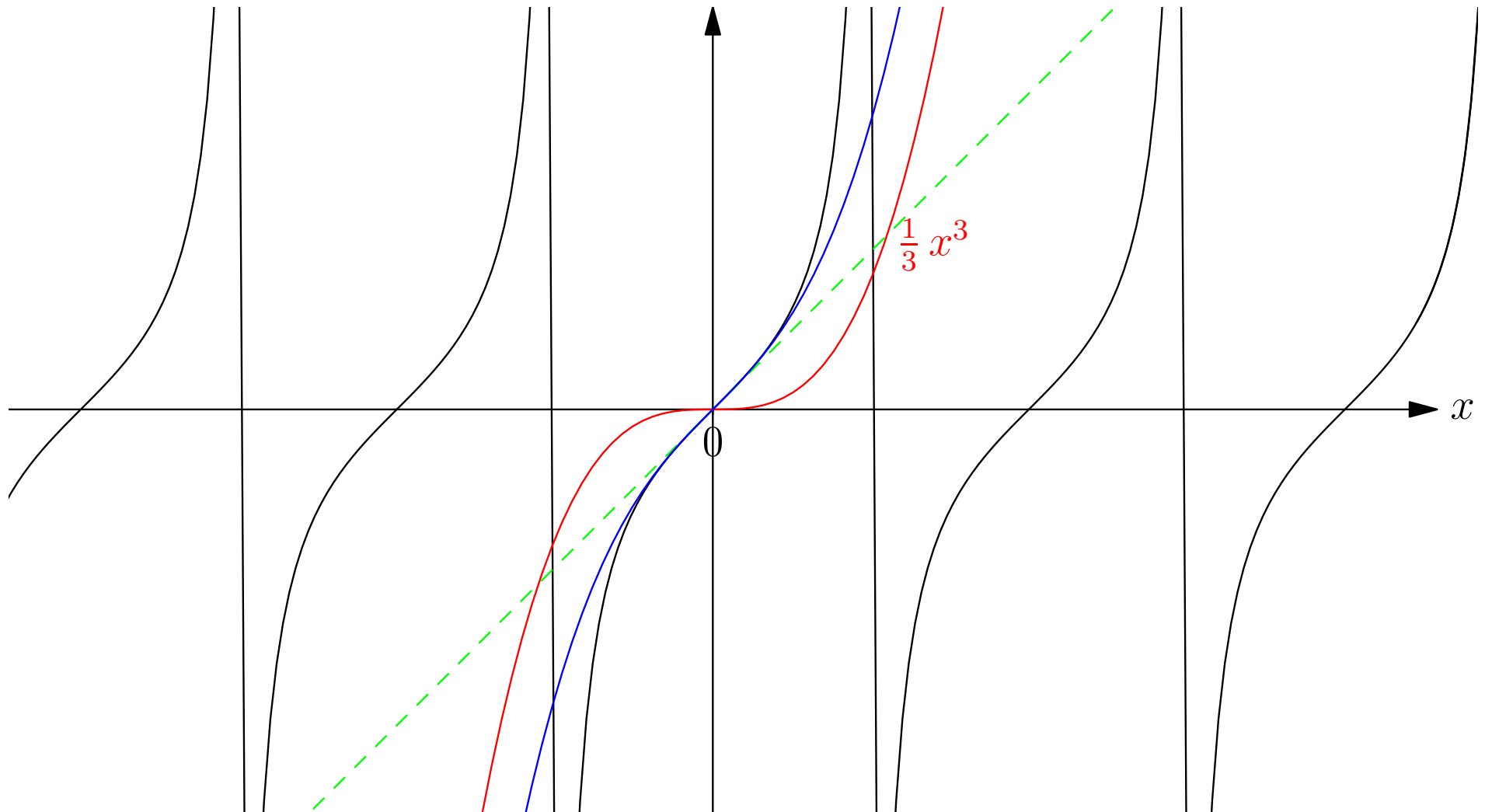
Taylor Expansion of $\tan(x)$



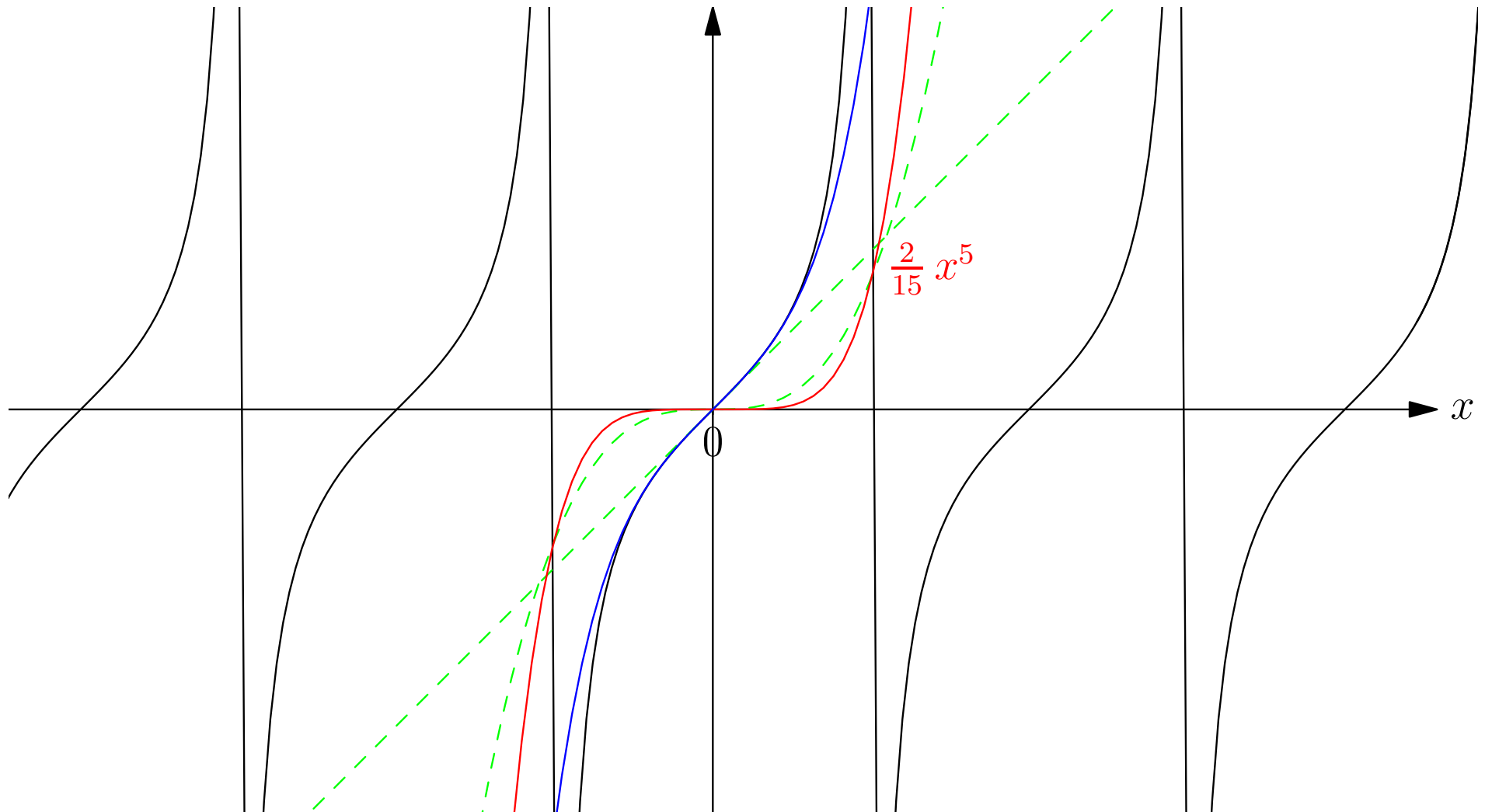
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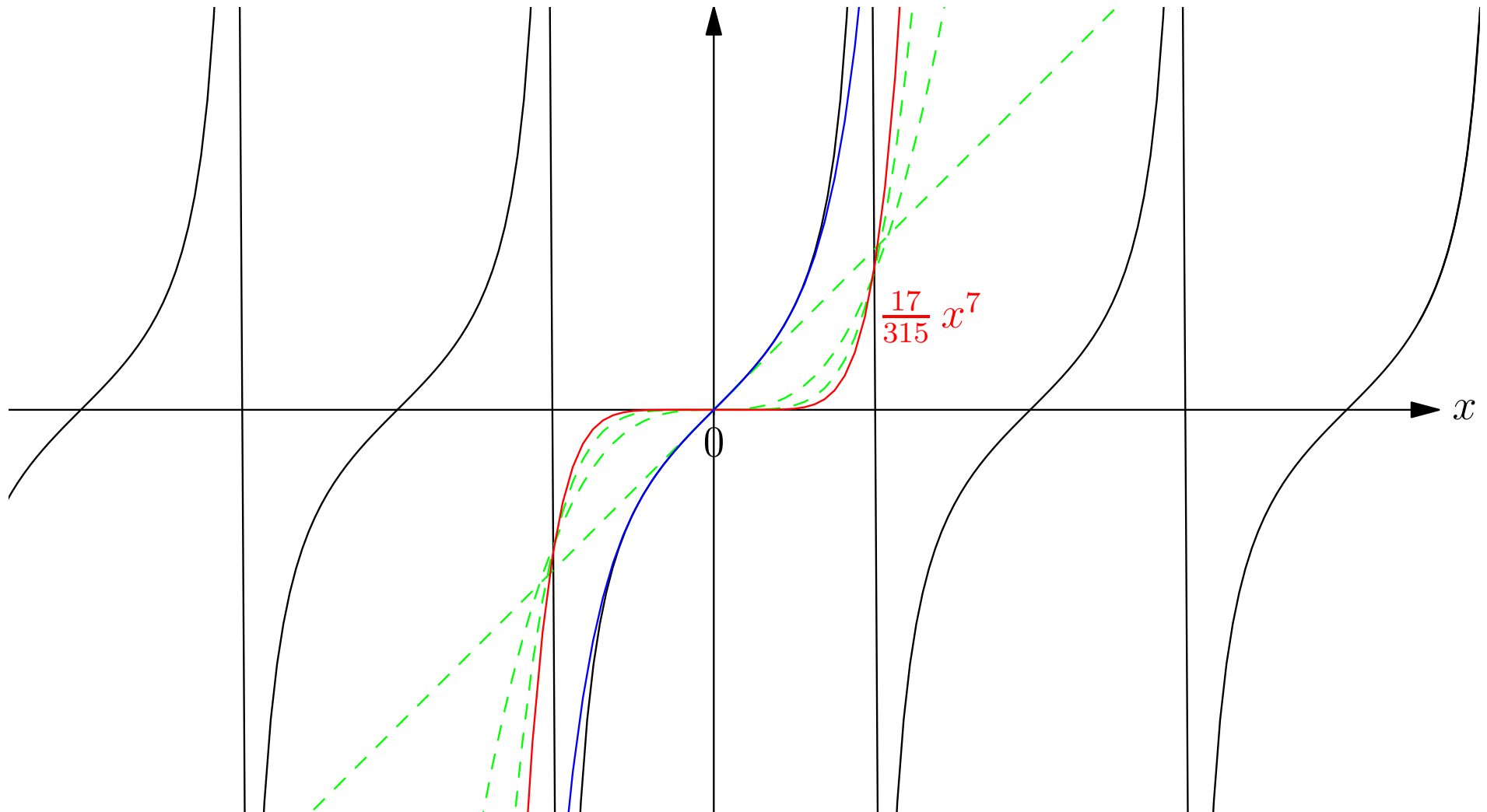
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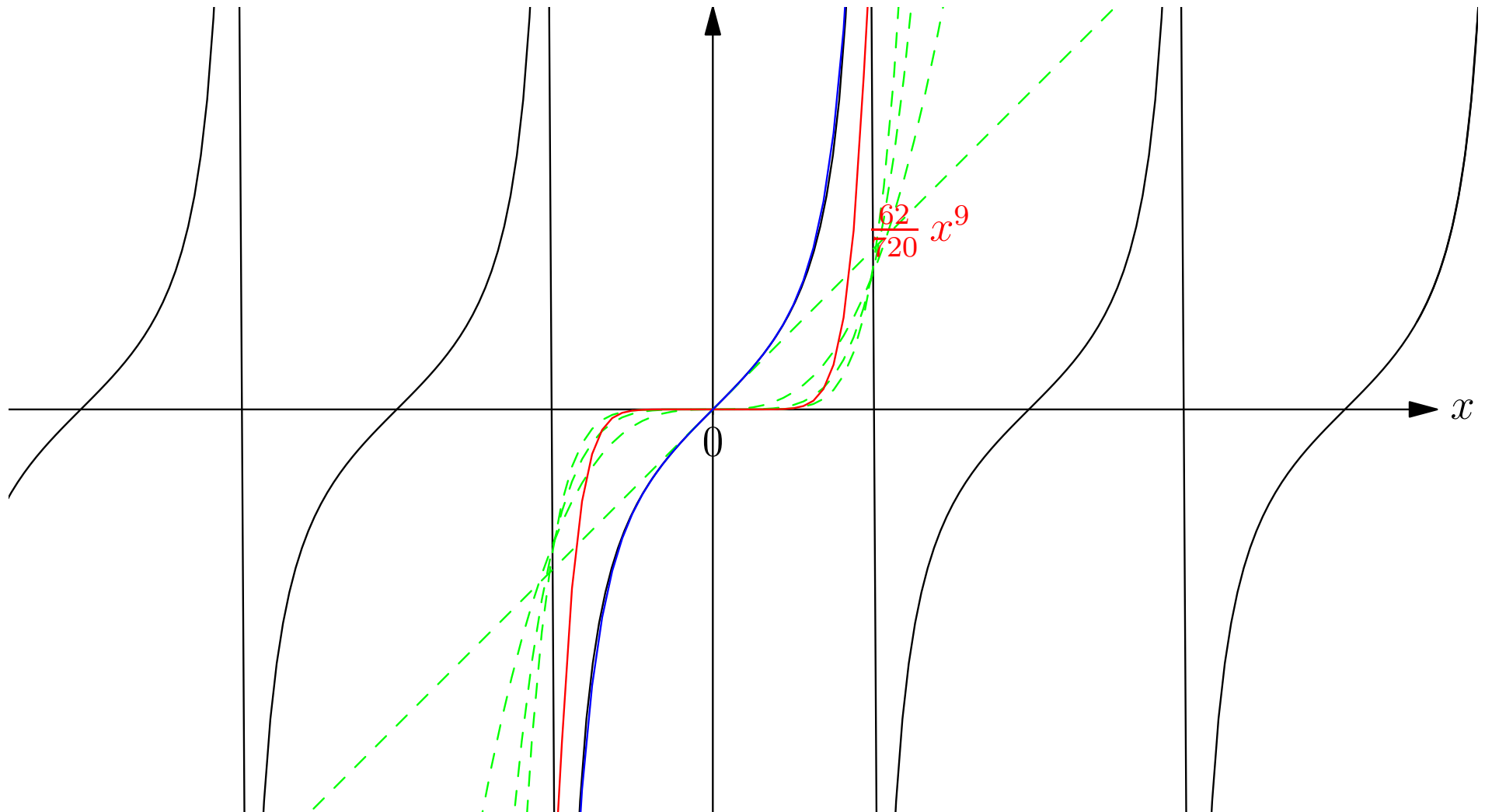
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Complex Functions

- To understand convergence of Taylor series more generally we need to go into the complex plane
- We can generalise many functions by considering them as a function with a complex argument $z = x + iy$
- The value of the complex function will be complex valued
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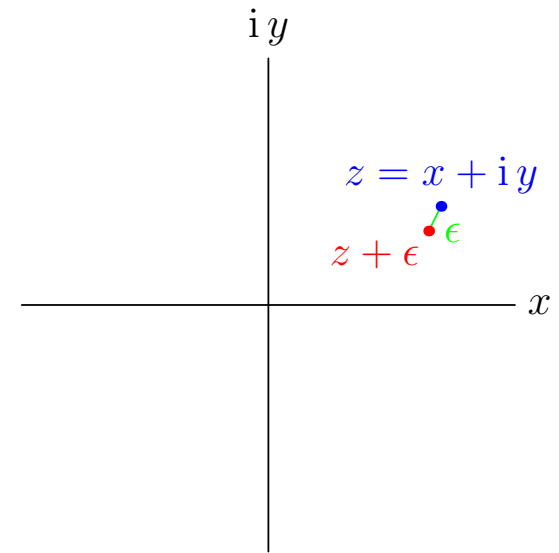
Analytic Functions

- An important class of functions are those where for almost all z the derivative

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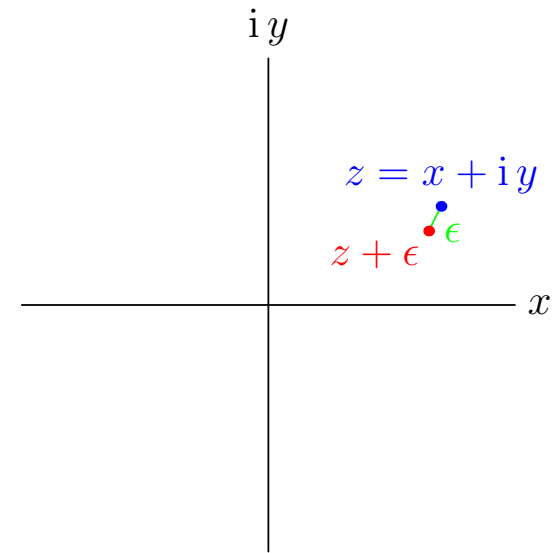


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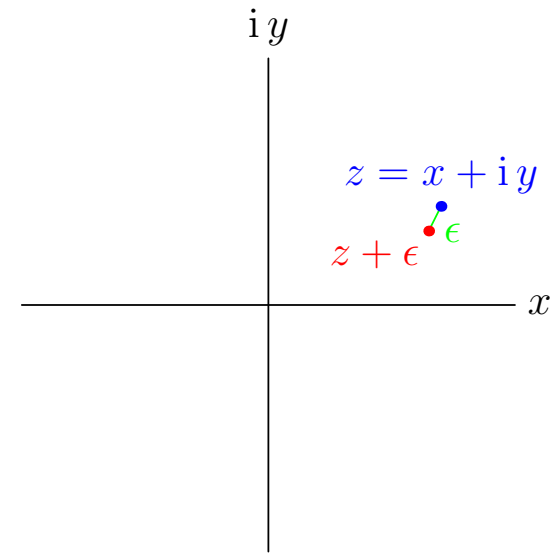
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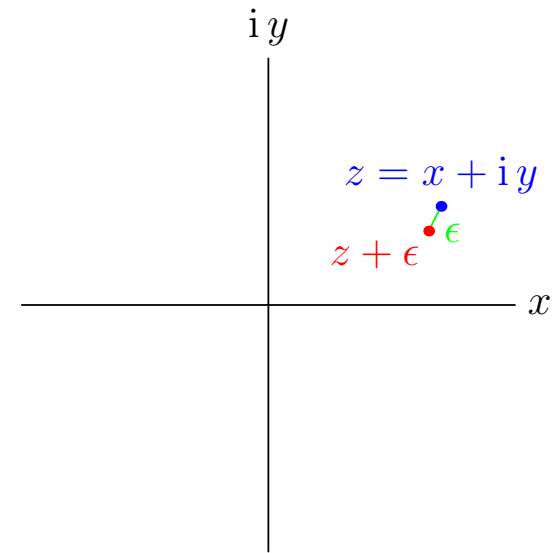


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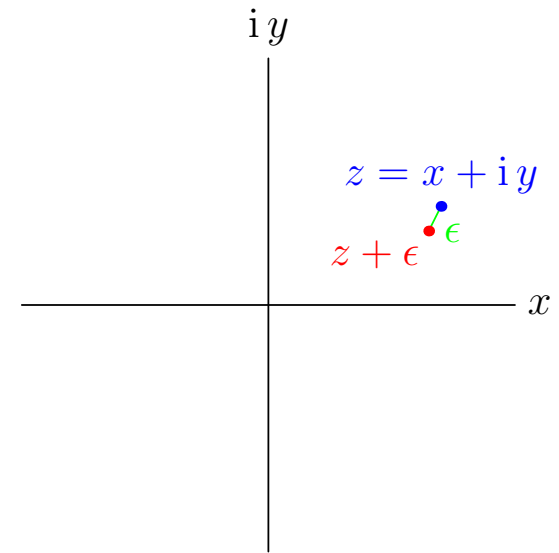
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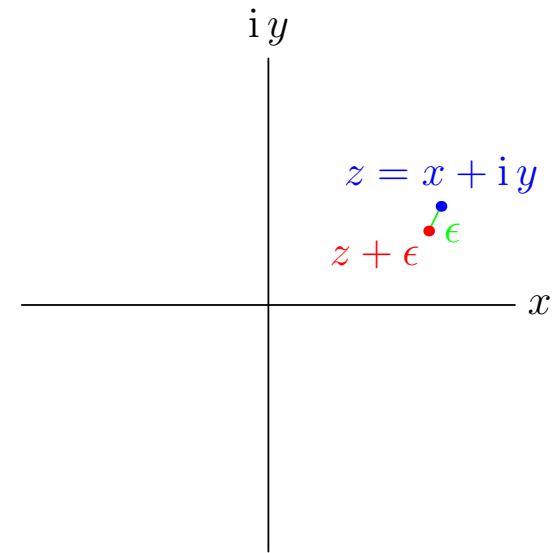
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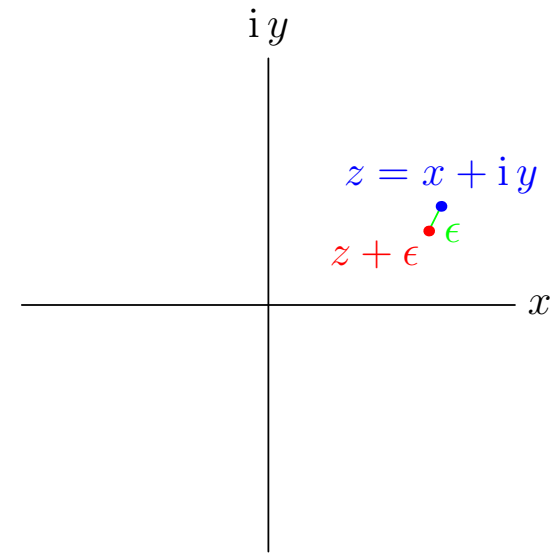
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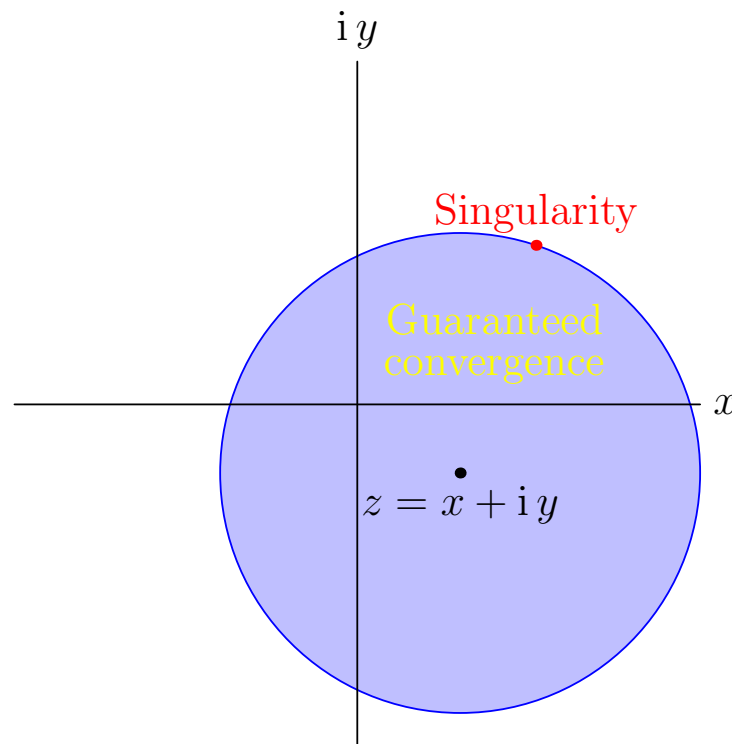


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Guaranteed Convergences of Taylor Functions

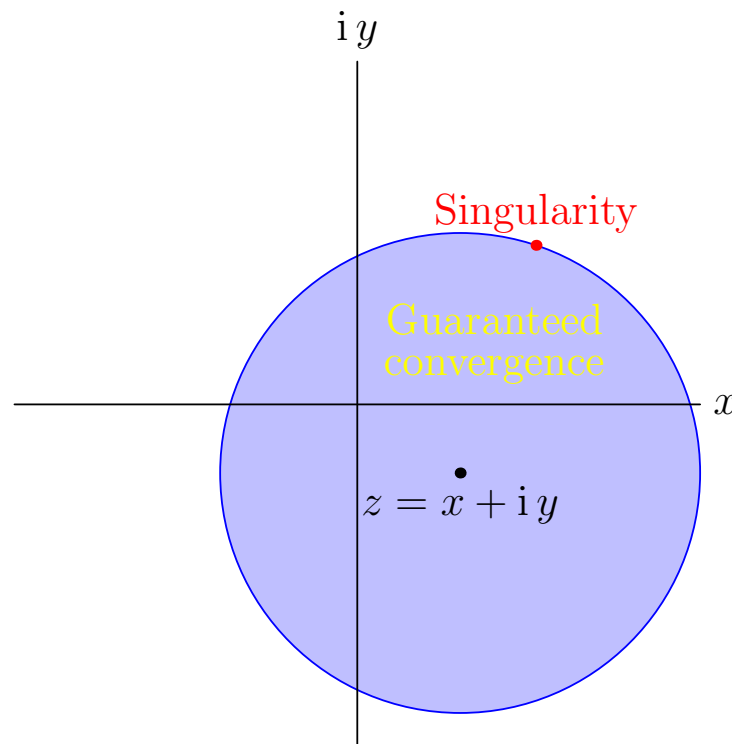
- For analytic functions, Taylor series are guaranteed to converge inside a circle that excludes singular points



- E.g. the function $f(z) = \frac{1}{(z+i)^2}$ has a singularity at $z = -i$

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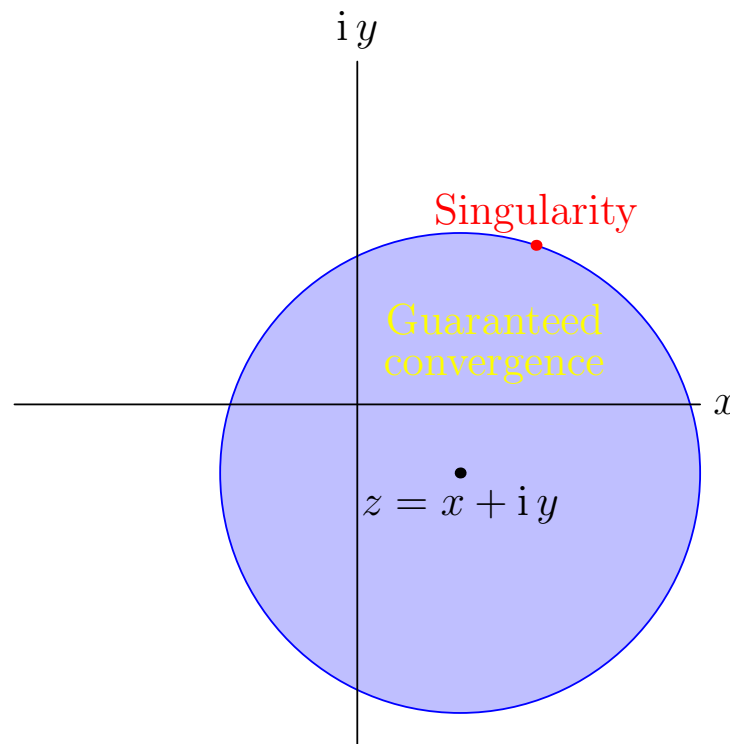
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Guaranteed Convergences of Taylor Functions

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- E.g. the function $f(z) = \frac{1}{(z+i)^2}$ has a singularity at $z = -i$. The Taylor series of $f(z)$ around 0 is guaranteed to converge inside a circle of radius 1

The Hyperbolic Tangent

- The hyperbolic tangent, $\tanh$, is related to the normal tangent function through $\tanh(x) = i \tan(ix)$
- The function (in the real plane) doesn't have any singularities
- But we know that $\tan(x)$ has a singularities at $\frac{\pi}{2} \pm k\pi$ where k is any integer
- Thus a Taylor expansion of $\tanh(x)$ around $x = 0$ is guaranteed to converge in a circle of radius $\pi/2$, but not necessarily for larger values of x

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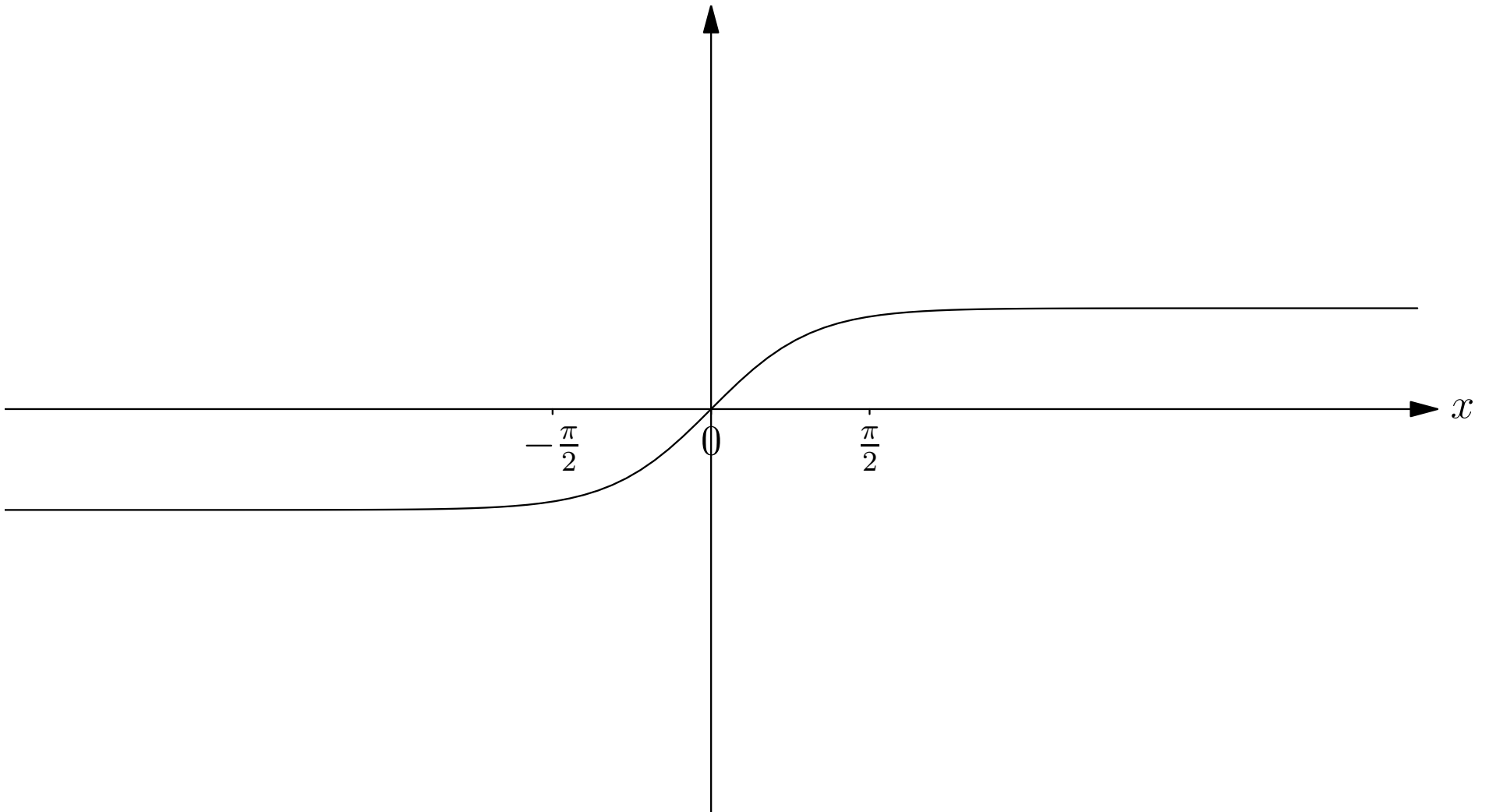
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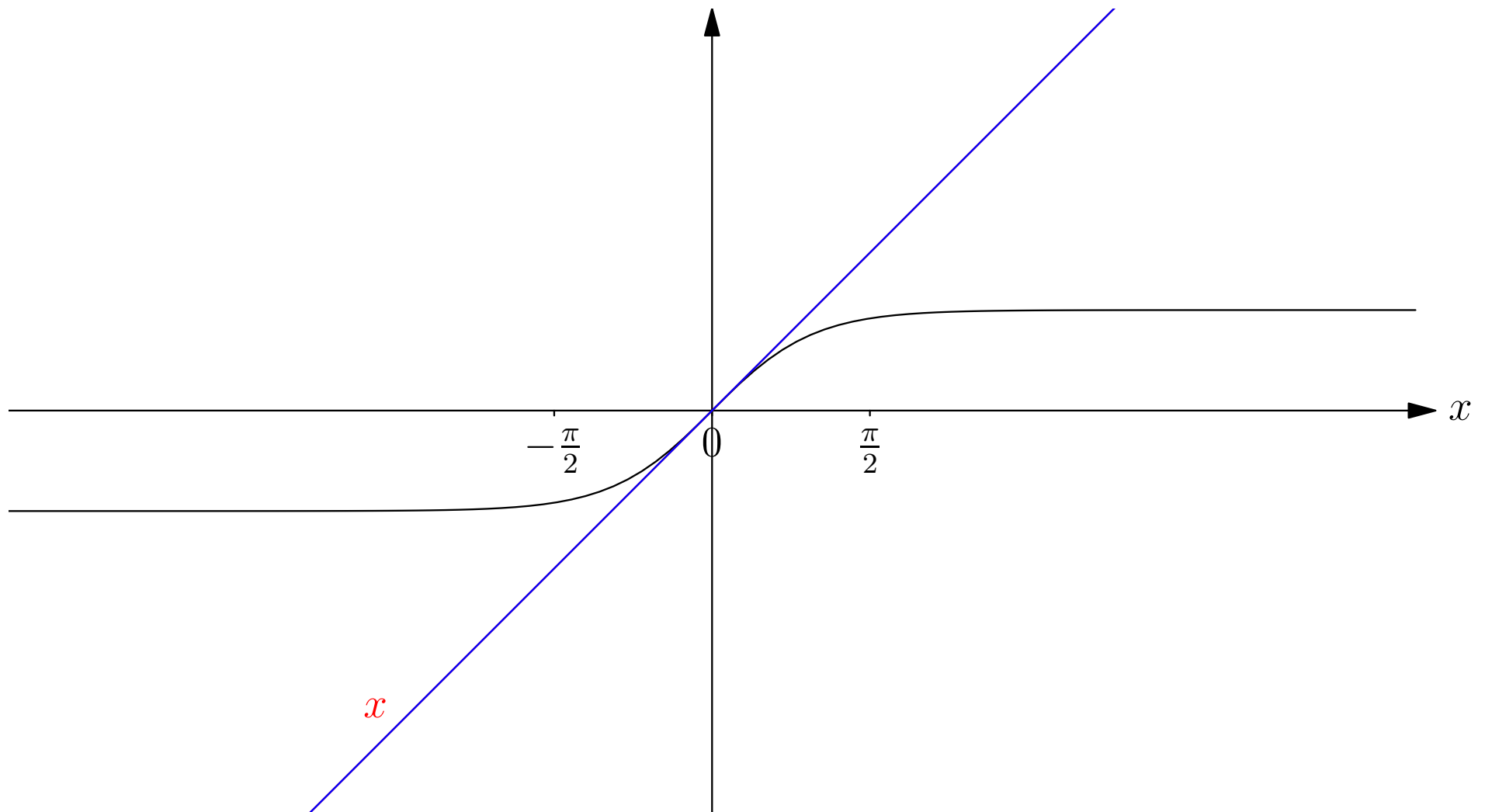
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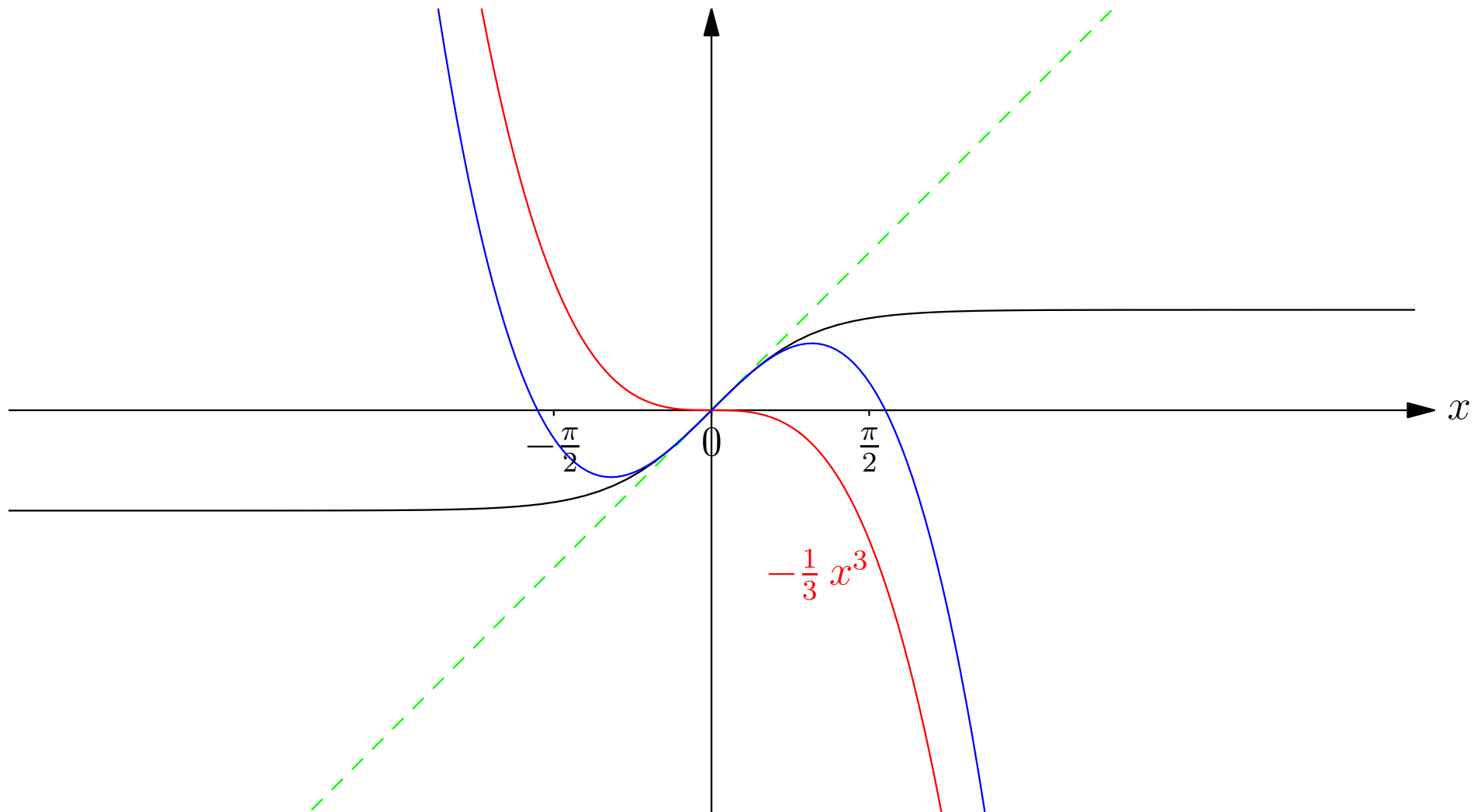
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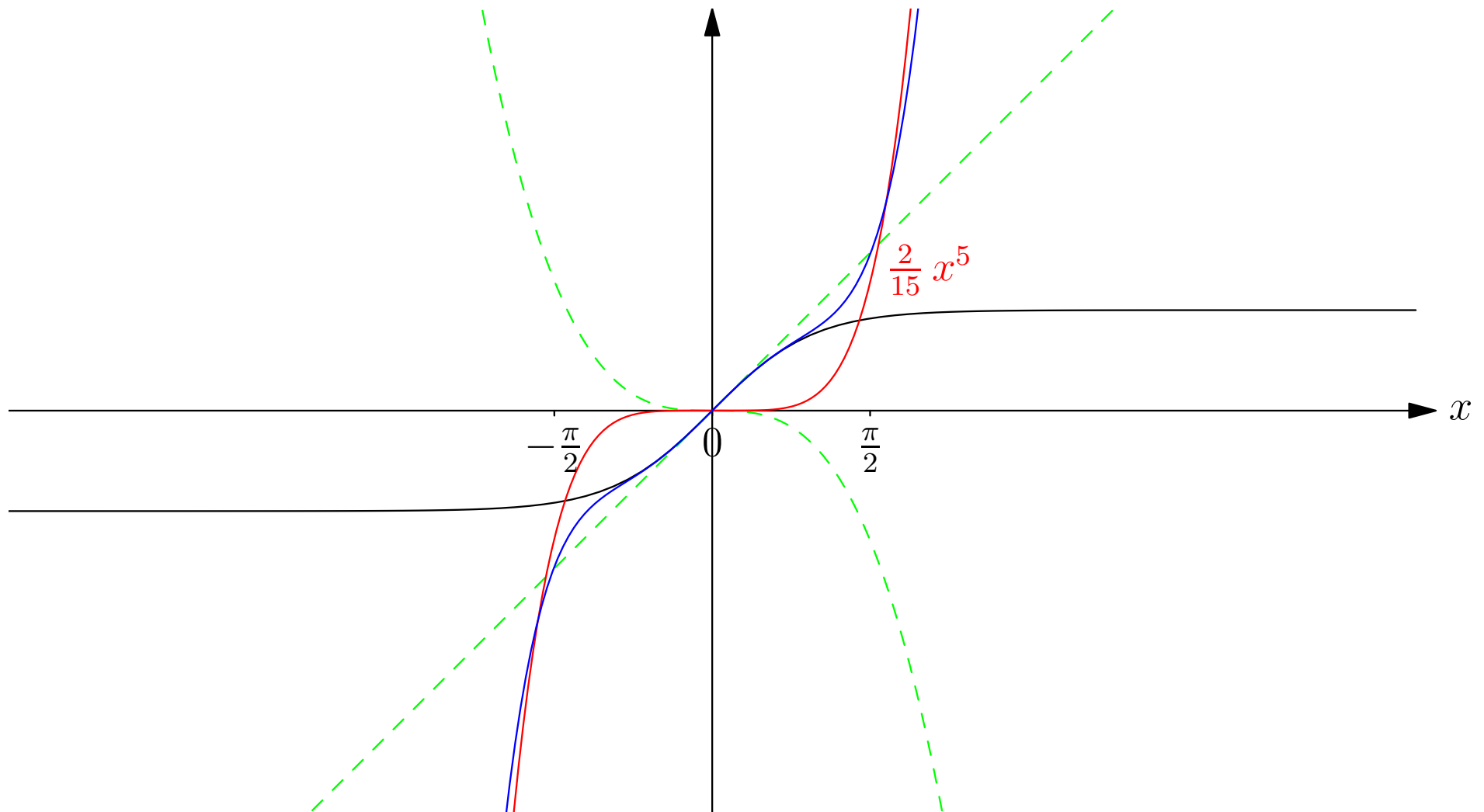
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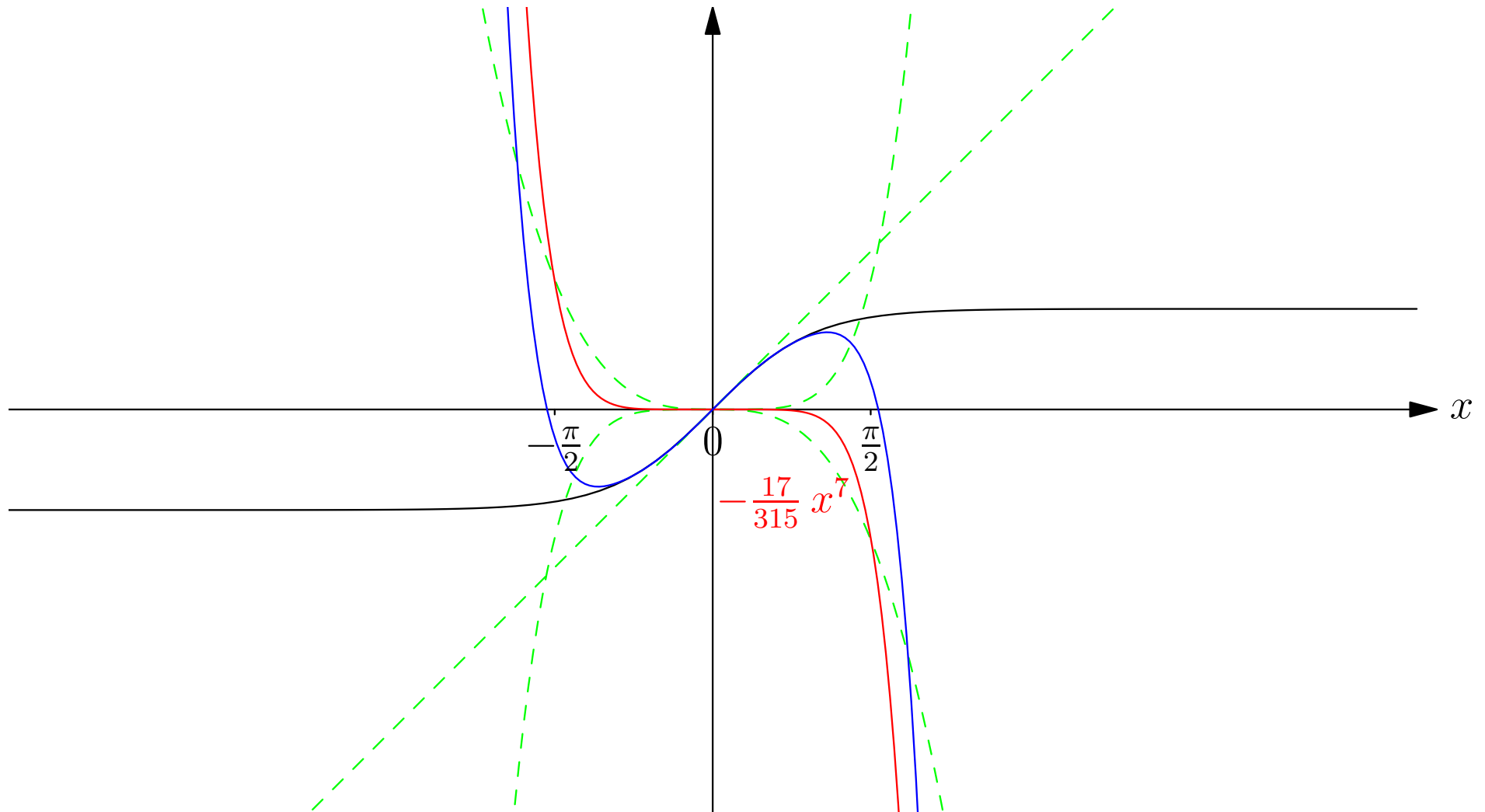
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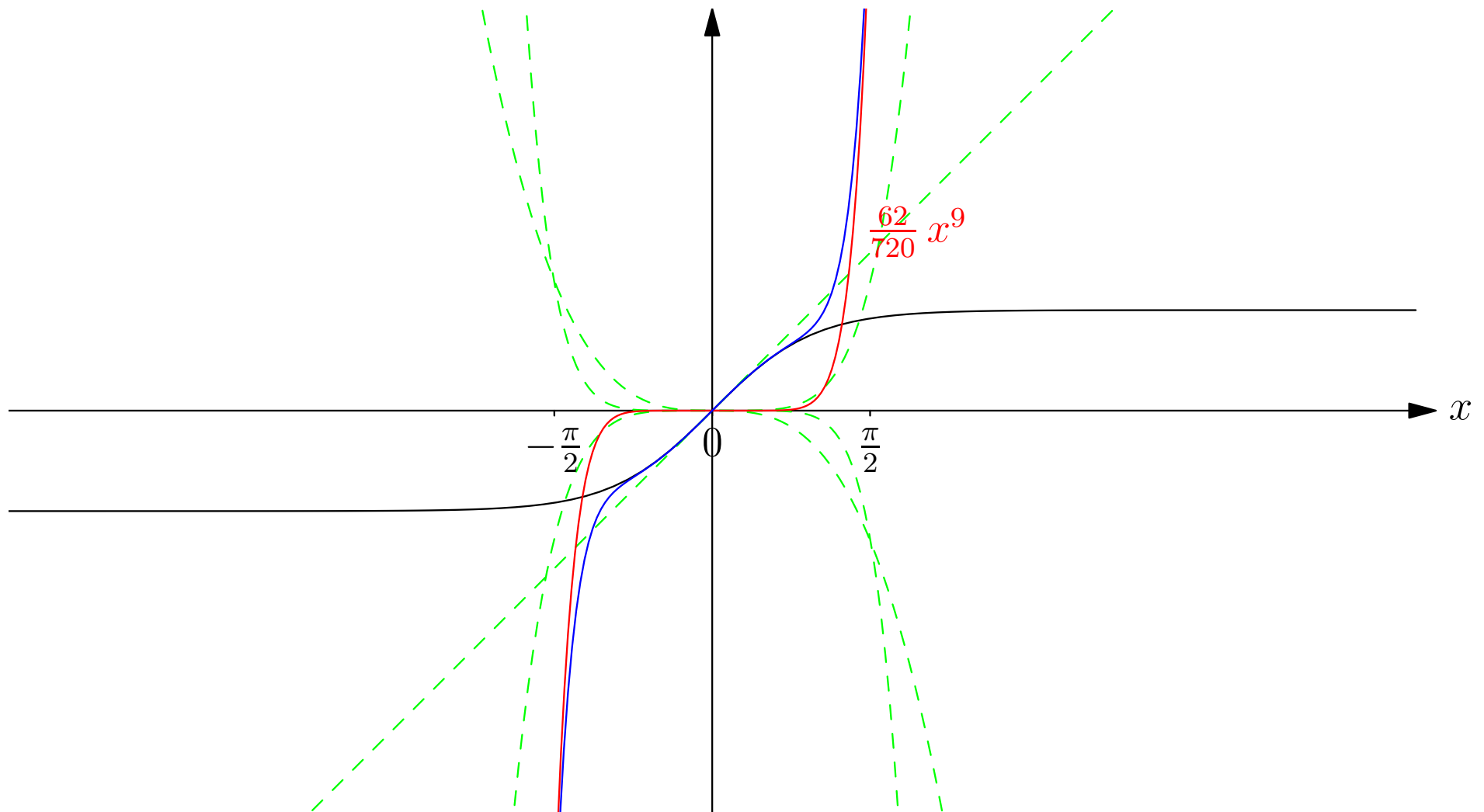
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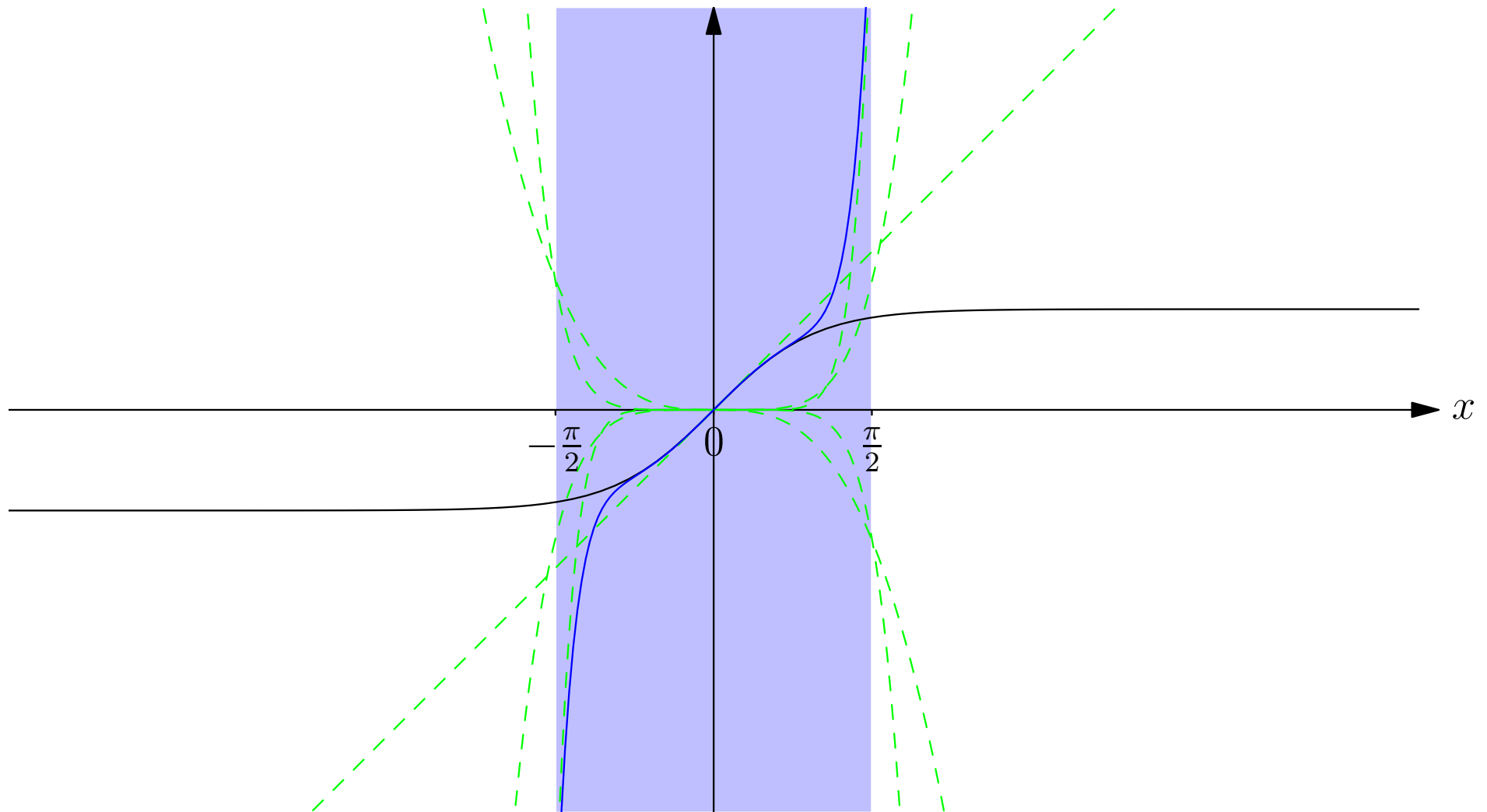
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Residue

- Taylor expansions actually come with an error bound

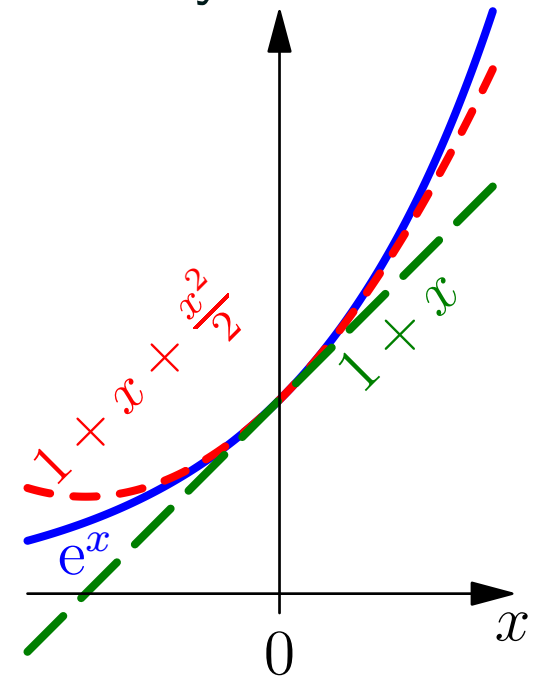
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θ_x is a number in the interval between x^* and x

- The last term known as the **residue** can often be very useful

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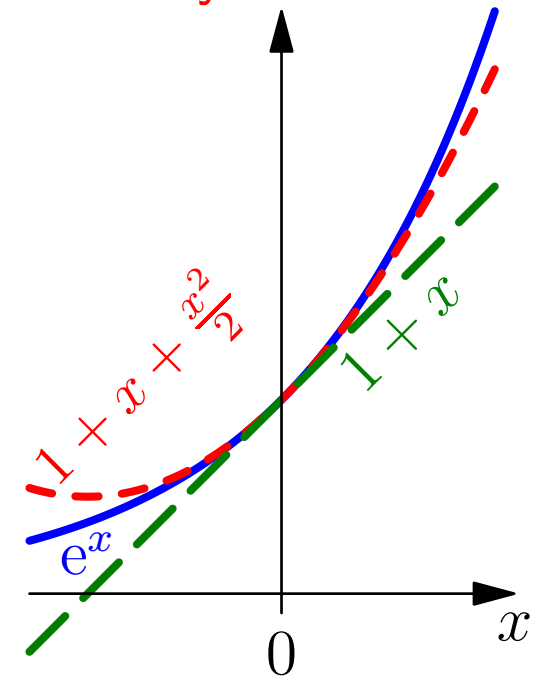
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Residue

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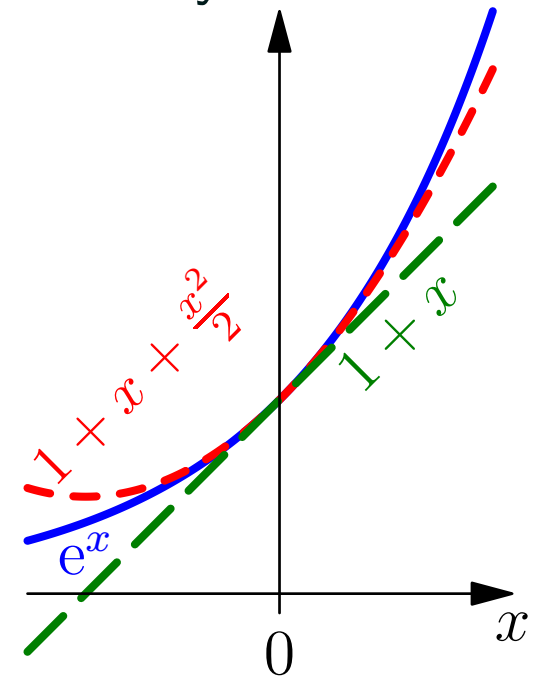
$$f(x) = \sum_{i=0}^{n-1} \frac{(x - x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x - x^*)^n}{n!} f^{(n)}(\theta_x)$$

θ_x is a number in the interval between x^* and x

- The last term known as the **residue** can often be very useful

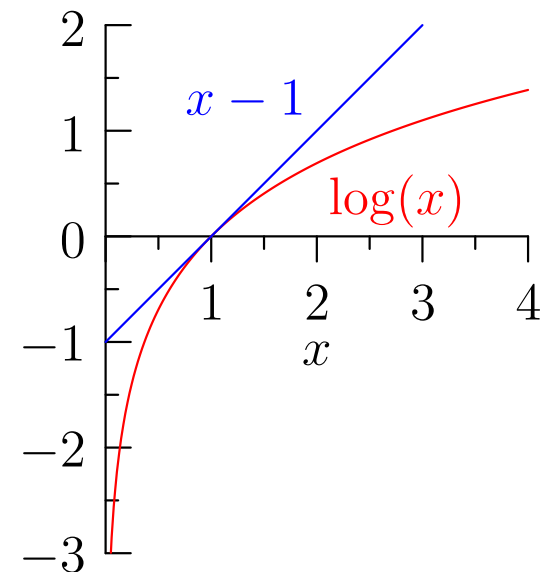
$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x}$$

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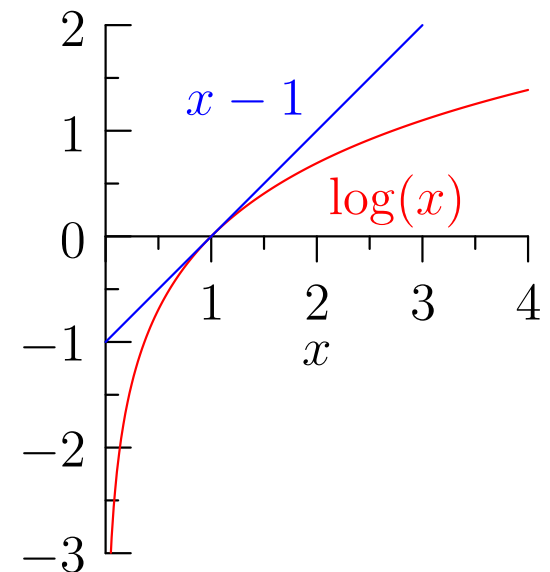
Useful Bounds

- A much used bound in mathematical analysis is $e^x \geq 1 + x$
- Many other bounds can be obtained similarly



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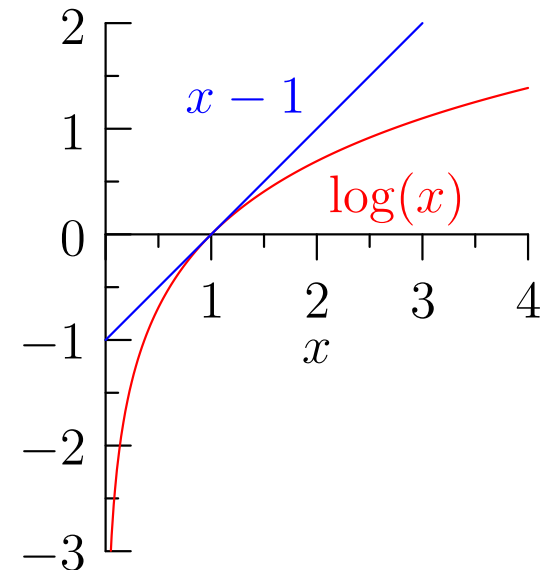
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$$\log(x) = (x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} - \dots$$



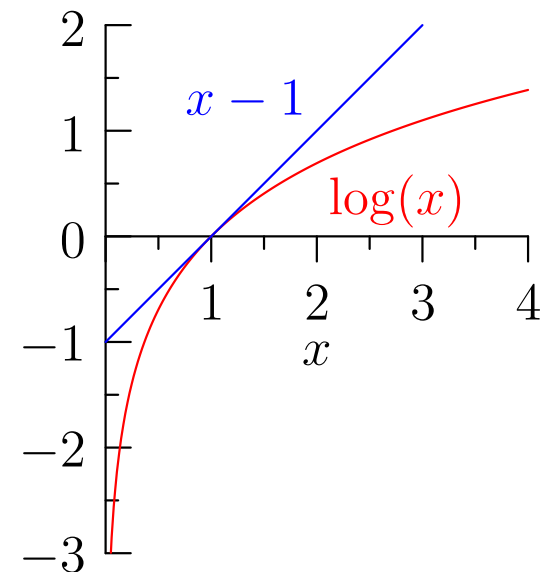
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giving

$$\log(x) \leq x - 1$$



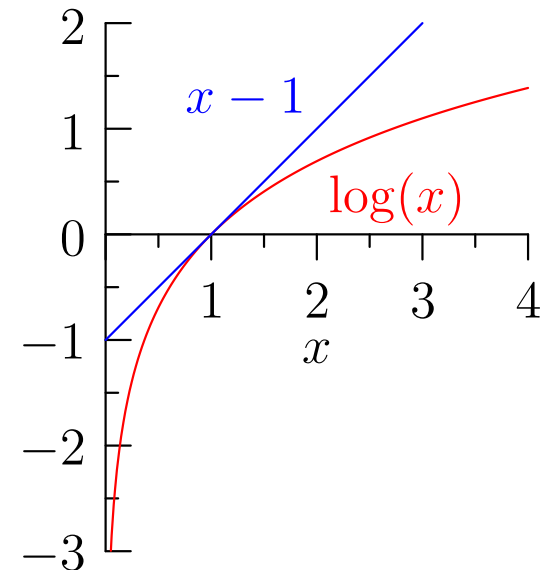
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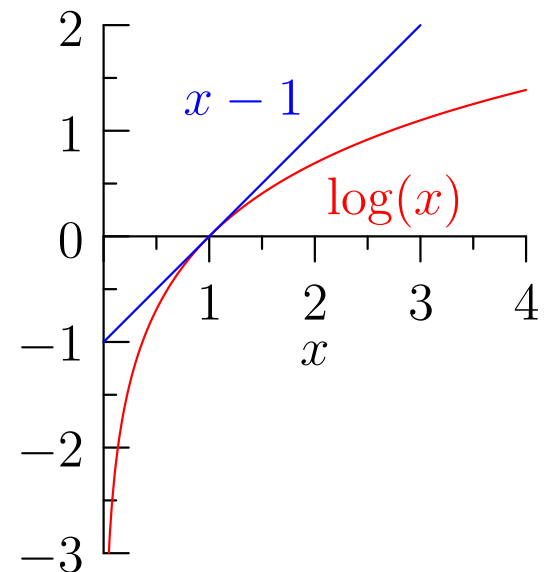
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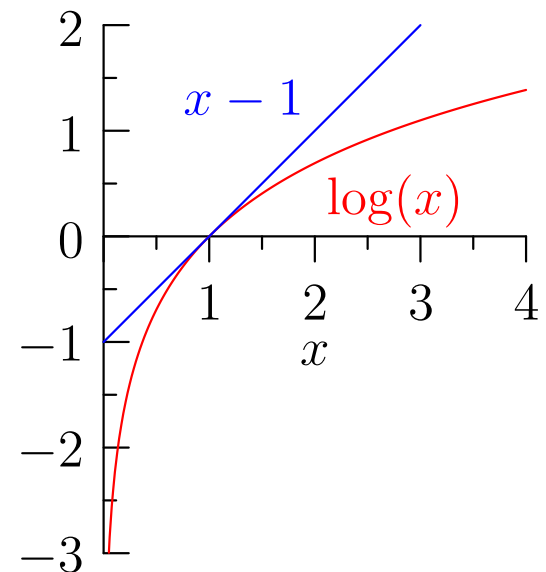
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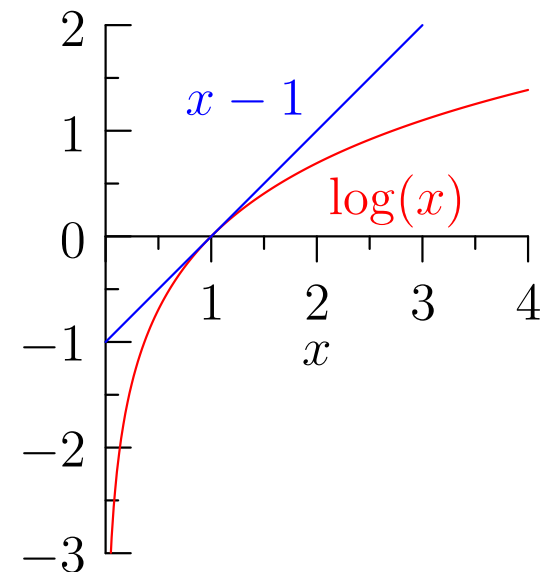
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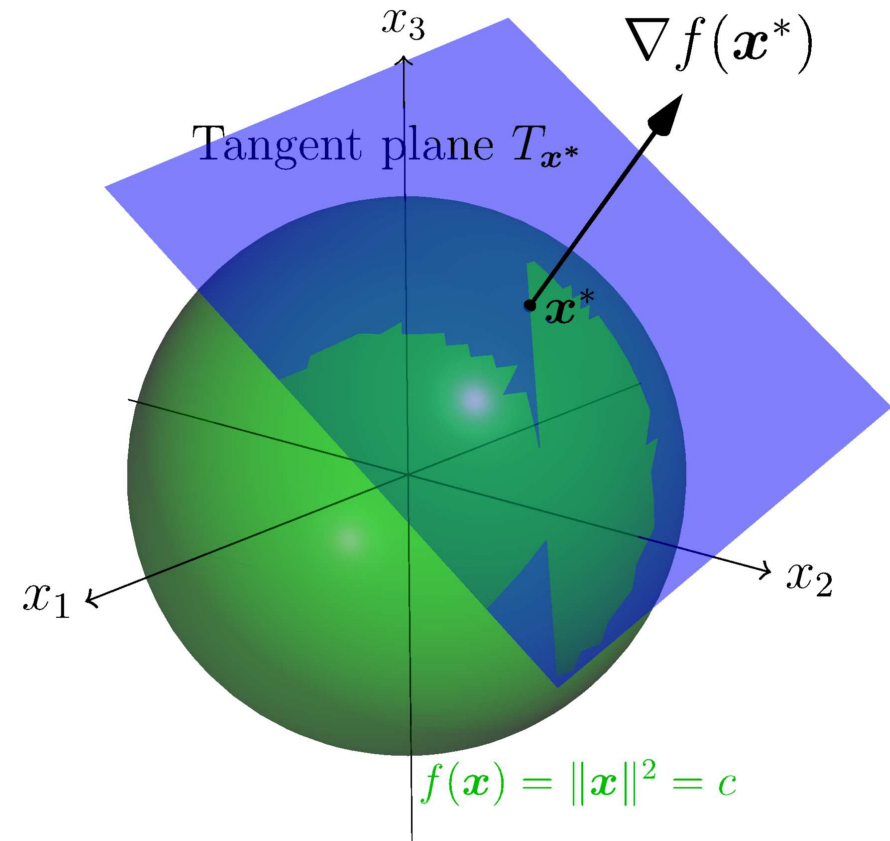
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- The proof of the residue for Taylors come from applying multiple integrals to the term $f^{(n)}(x)$. For the sake of time I will skip this

Outline

1. 1-D Taylor Expansion
2. **High Dimensions**
3. Hessians



High Dimensional Spaces

- Just as we defined a Taylor expansion by approximating the function to a polynomial we can do the same for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\mathbf{x}) \approx c^{(0)} + \sum_{i=1}^n c_i^{(1)} (x_i - x_i^*) + \sum_{i,j=1}^n (x_i - x_i^*) c_{ij}^{(2)} (x_j - x_j^*) + \dots$$

- We can determine the coefficients by differentiating both sides and setting $\mathbf{x} = \mathbf{x}^*$

$$f(\mathbf{x}^*) = c^{(0)}$$

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$$\frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j} = 2c_{ij}^{(2)}$$

High Dimensional Taylor Expansion

- We find

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + \dots$$

- Where the matrix $\mathbf{H}$ is known as the Hessian and has components

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

- Because

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

the Hessian is symmetric

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the Hessian is symmetric ($\mathbf{H} = \mathbf{H}^\top$)

Tangent Plane

- For smooth functions in n dimensions the equation $f(\mathbf{x}) = c$ defines a $(n - 1)$ -dimensional hyper-surface
- This can be thought of as the high dimensional analogue of a contour line
- The plane that goes through a point $\mathbf{x}^*$ that lines up with the contour surface is known as the tangent plane
- Away from $\mathbf{x}^*$ the value of the function $f(\mathbf{x})$ differs from the tangent plane by a quantity of order $\|\mathbf{x} - \mathbf{x}^*\|^2$

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Defining the Tangent Plane

- Thus if $\mathbf{x}$ is in the tangent plane $T_{\mathbf{x}^*}$ then

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + O((\mathbf{x} - \mathbf{x}^*)^2)$$

- Or $(\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) = 0$
- That is the tangent plane at a point $\mathbf{x}^*$ is the set of points orthogonal to the gradient
- Notice that the gradient then points in the direction where the function increases fastest

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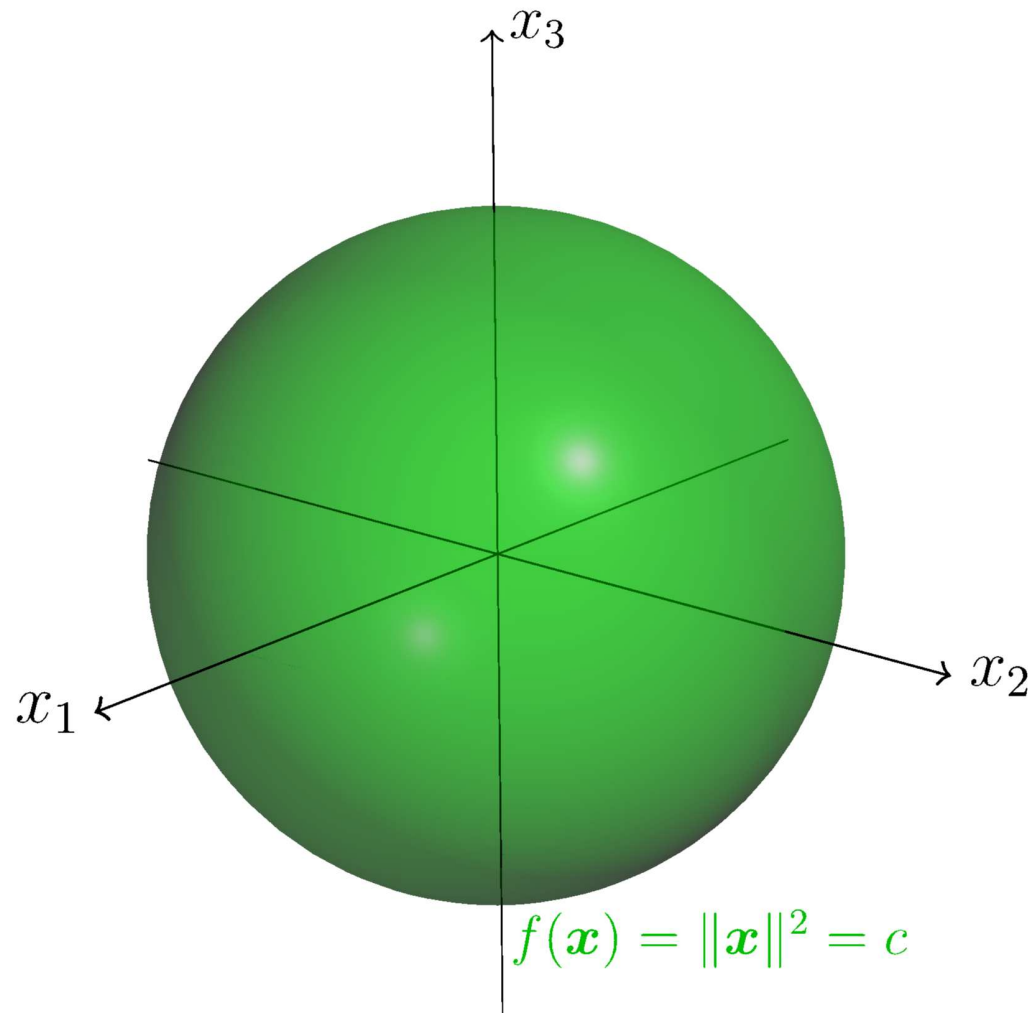
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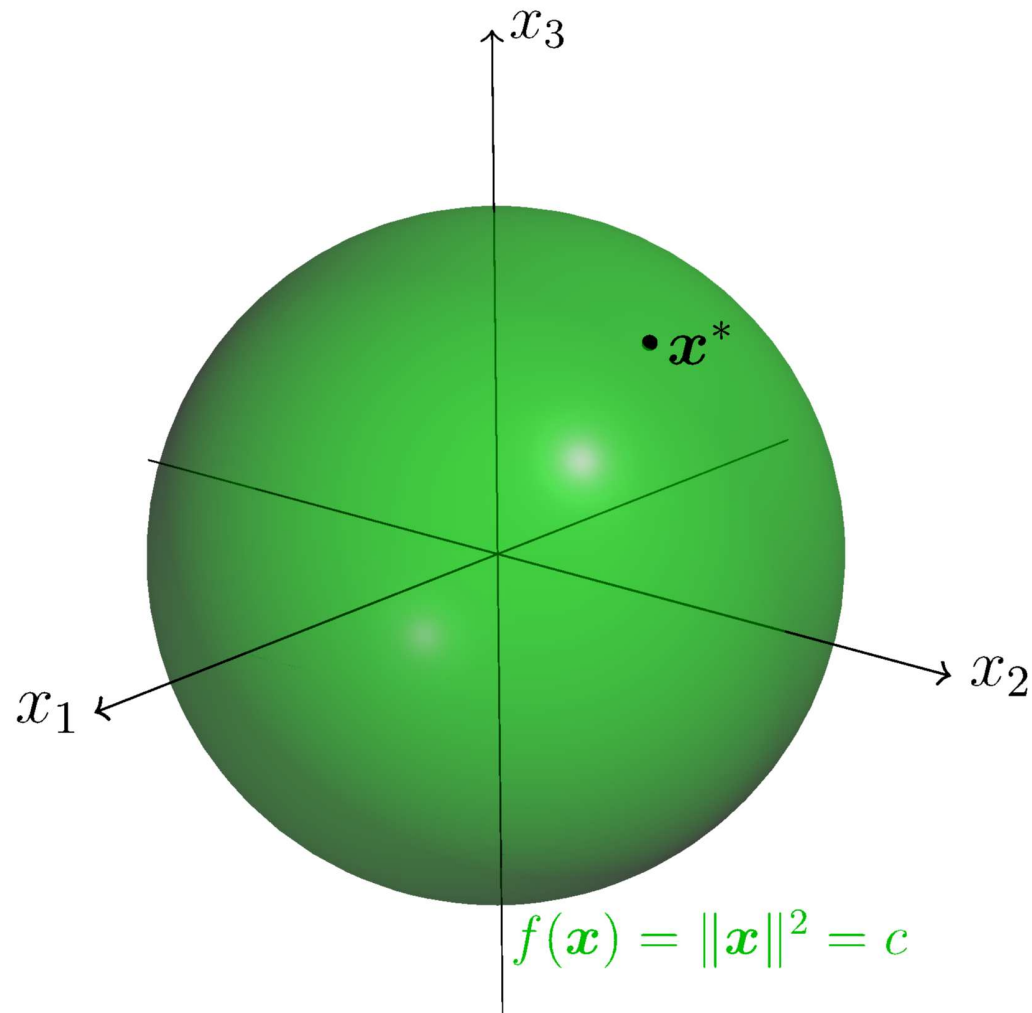
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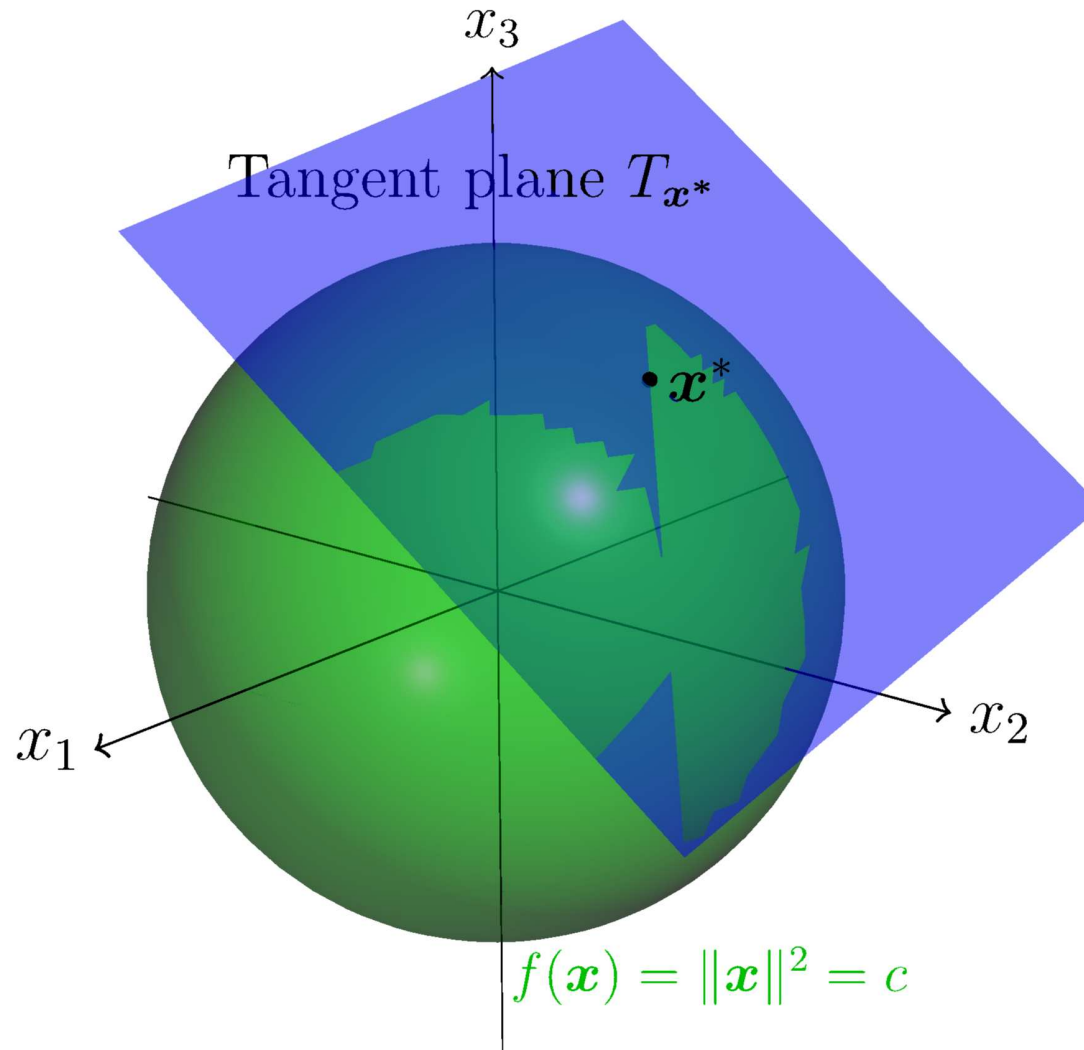
Picture of Tangent Plane for $f(\mathbf{x}) = \|\mathbf{x}\|^2 = c$



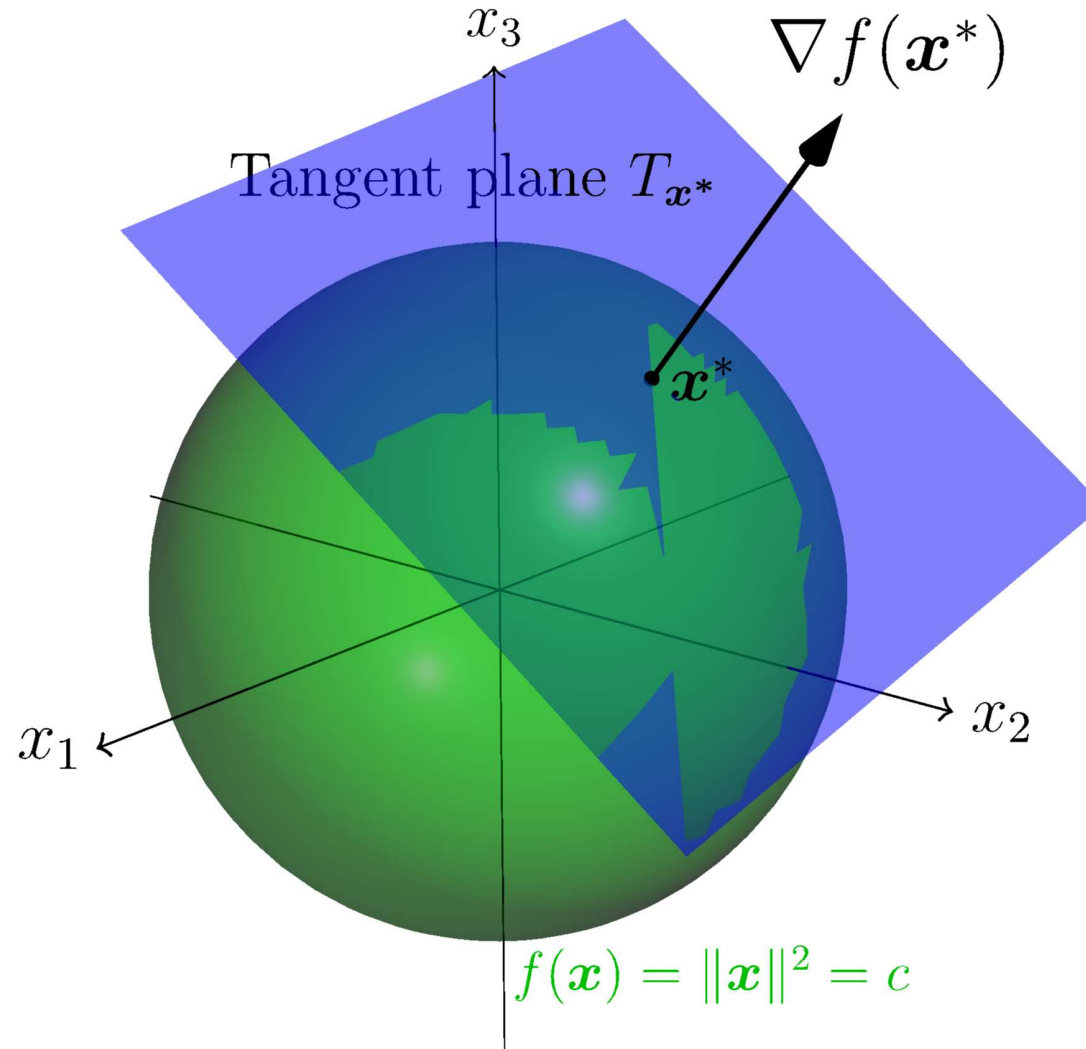
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Direction of Steepest Ascent

- We note that as

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}^*) + O(\epsilon^2)$$

- The direction where local the function grows fastest is when $\mathbf{u}$ is in the direction $\nabla f(\mathbf{x}^*)$
- Likewise when $\mathbf{u}$ is in the direction $-\nabla f(\mathbf{x}^*)$ the function decreases as fast as possible
- This is only true locally (when $\epsilon \gg \epsilon^2$)

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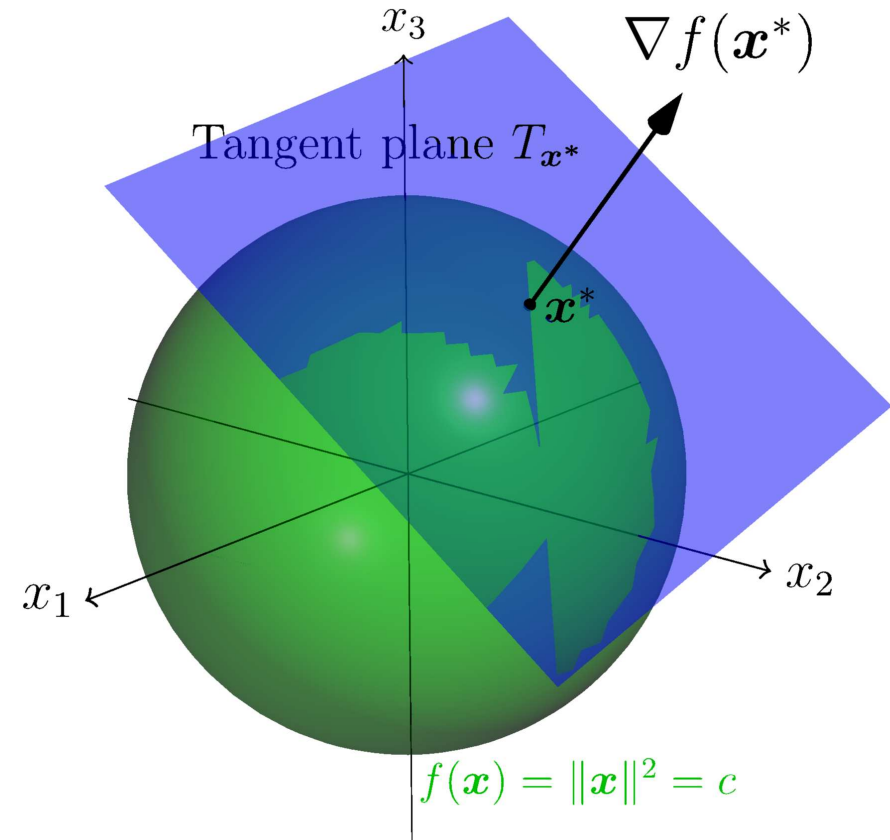
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3. **Hessians**



The Hessian

- The second order term in the Taylor expansion tells us about the **curvature** of the function
- This is very relevant when we are at an **extremal point**, $\mathbf{x}^*$, where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ so that

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^3)$$

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- For sufficiently small perturbation the sign of change in the function depends on the sign of $\mathbf{u}^\top \mathbf{H} \mathbf{u}$

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Eigenvalues

- We will revisit eigen-systems, but recall that for a square matrix an eigenvector, $\mathbf{v}_i$, of a matrix $\mathbf{H}$ is defined such that

$$\mathbf{H}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- For an $n \times n$ symmetric matrices (such as the Hessian), the eigenvalues are all real, and we can choose n orthogonal eigenvectors such that $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$
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- For an $n \times n$ symmetric matrices (such as the Hessian), the eigenvalues are all real, and we can choose n orthogonal eigenvectors such that $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij} = \mathbb{I}[i = j]$
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- We can write any vector $\mathbf{u}$ as a weighted sum of the eigenvectors (the eigenvectors form a basis) $\mathbf{u} = \sum_{i=1}^n c_i \mathbf{v}_i$
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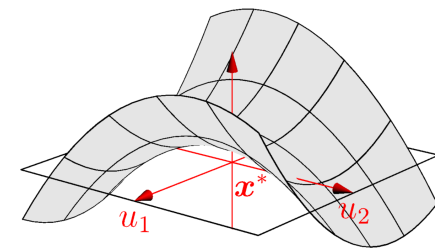
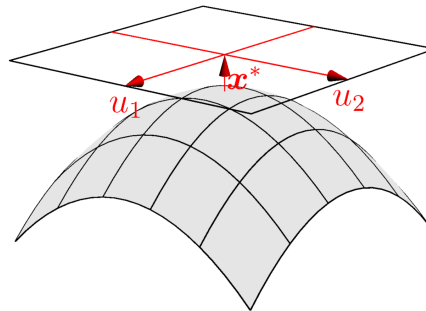
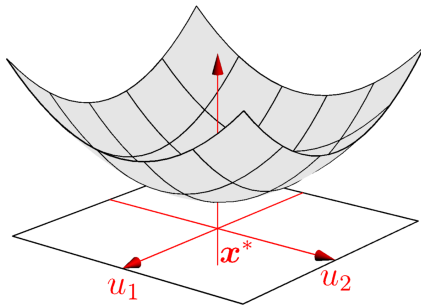
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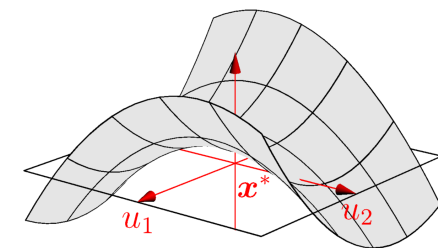
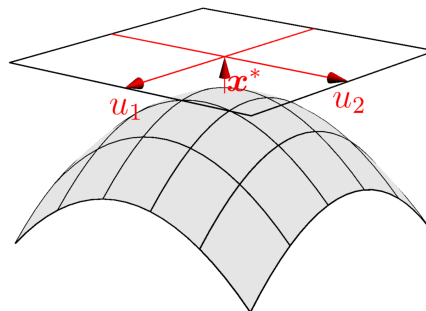
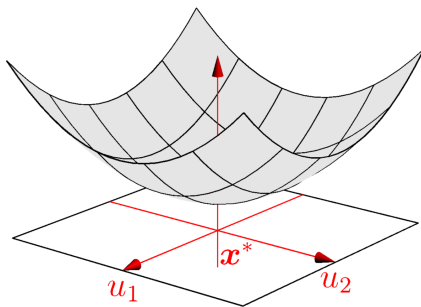
Minima, Maxima and Saddle-Points

- If all the eigenvalues are positive, $\lambda_i > 0$, then for any u we have $u^T H u > 0$ and x^* would be a local minimum
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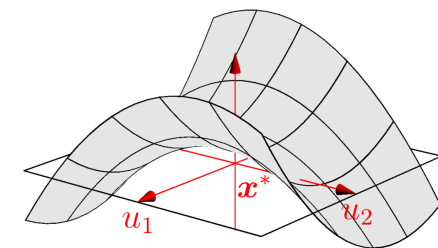
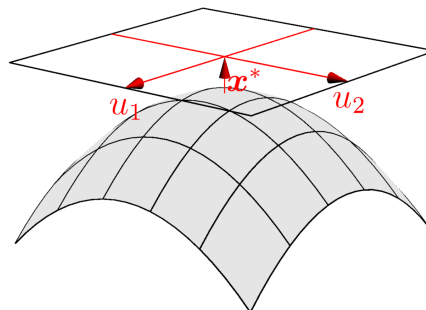
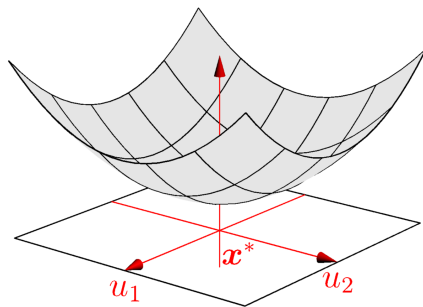
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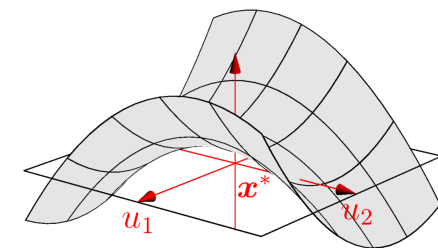
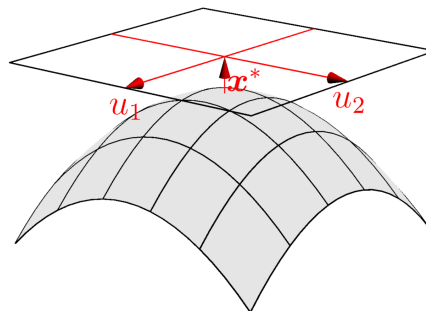
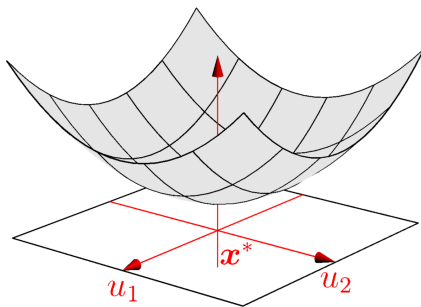
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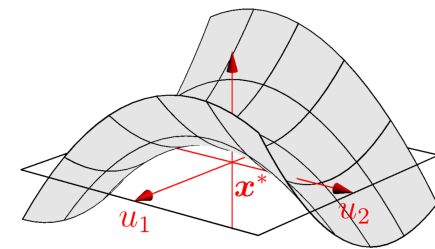
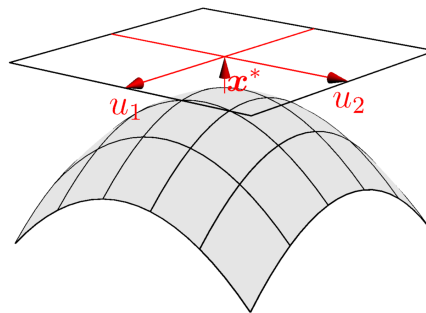
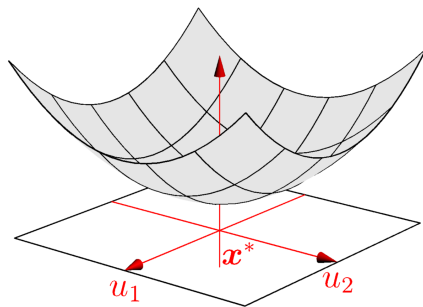
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Finding Optima

- We are often given a loss function, $L(\mathbf{w})$, which we wish to minimise
- The condition for $\mathbf{w}^*$ to be a minimum is that $\nabla L(\mathbf{w}^*) = \mathbf{0}$
- Except for a few very simple loss functions, there are likely to be many (and often a huge number) of local minima
- Unless $L(\mathbf{w})$ is very simple we are unlikely to be able to solve $\nabla L(\mathbf{w}) = \mathbf{0}$ analytically

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- We usually proceed iteratively, starting from some $\mathbf{w}^{(0)}$

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - r \nabla L(\mathbf{w}^{(t)})$$

- This moves us in the direction that (locally) minimises the loss function $L(\mathbf{w})$
- As we get closer to the minimum $\nabla L(\mathbf{w}^{(t)})$ typically becomes smaller and we take smaller steps
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- The gradient defines the direction in which a function increases fastest
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